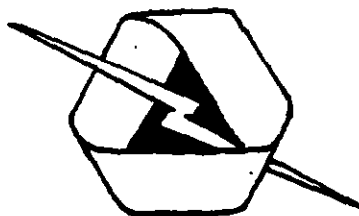


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TITLE:

ANTI-GRAVITY
ARTICLES # 2

013-AA2 ANTIGRAVITY ARTICLES (#2) --- Inspiring new & old information of useful and theoretical sorts including D. Buehler: Movement of High Voltage Dielectrics (2% Weight loss!)..Fischbach, et al.:Hypercharge 5th Force that counteracts Gravity..W.J. Hooper: Motional Electr. Field Theory ..R. Dratch (1979): E & H Vector Anti-G.. Dr. R.L. Clark: Philadelphia Expt. Explained..Ryan Aero. Co. Force Field Research..Biefeld-Brown Effect details (T.T. Brown)..R. Forward: "Guidelines To Antigravity" (3 Non-Newtonian gravitational forces described & recommendations..Tesla Space Drive..20 Articles.. 45 pp.

these facts fly in the face of "conventional wisdom." However, these observations are exactly what I have been for years and are explicitly depicted on page 143 of my book, "THE ENERGY MACHINE OF JOSEPH NEWMAN."

FACT 39: To provide even more depth to my Pioneering Technology I gave a demonstration of how my technology immediately and realistically be applied to space travel.

my

MOBILE REGISTER



Press Register photo by Jay Perchard

Joe Newman, Lucedale inventor, watches his mini-spaceship rise in a demonstration to an audience at the Admiral Semmes Hotel in Mobile Wednesday morning.

Newman unveils new invention

By JUNE HEYDT

Press Register Correspondent

Joseph W. Newman, the Lucedale inventor who claims he has invented an energy machine which produces more energy than it consumes, Wednesday demonstrated still another invention, this one a mini-spaceship, that he said uses this same technology.

Newman said his mini-spaceship employs his revolutionary concept of electromagnetism and the gyroscopic atoms.

"I have on numerous occasions stated that within 20 years space travel will be as commonplace as getting on a plane and flying from one city to another by the use of the pio-

neering technology that I teach," Newman said.

"Space travel can become a safe reality now—not be a potential tomb for good people by placing them in a tremendously inefficient liquid bomb, which makes the potential of a large bomb with cramped living conditions for its brave and courageous occupants," Newman said at the Admiral Semmes Hotel in Mobile.

"My technology can build space devices having dimensions of immense area thereby giving means to carry a great cargo to the extent that a miniature world environment

■ Please see Page 10, Col. 4

Newman

■ Continued From Page 1

could be transported having great beauty and practicality and giving great hope and excitement because the mysteries of the universe will be open to true explorations and inhabitation."

During the demonstration, Newman explained to the audience that he could take a coil of wire and create a magnetic field equal to the city of Mobile.

Dr. Roger Hastings, a prominent physicist, who was also at the demonstration said, "I have recently witnessed a demonstration of Newman's electromagnetic air and space vehicle demonstration prototype. It consists of an aluminized helium balloon wrapped with No. 38 gauge copper wire. The system is nominally heavier than air. When the wire coil is connected to a 200-volt battery, the balloon gradually lifts into the air. It aligns with the earth's magnetic field as it rises. If the

current is cut off, the balloon immediately begins to fall. If the current direction is reversed, the balloon rotates to align with the earth's field."

Newman said, "By simply changing the magnet you can make it go anywhere you want it to go. I'm certain this device (mini-spaceship) will reach the speed of light. What you're seeing has never been done before in the history of the world."

"In five years this technology could be in space," Newman said. "I don't just blow smoke; that's a fact. I can do things the scientific community says are impossible."

Hastings explained, "Newman's design calls for creation of a magnetic field which is an average about equal to the earth's field over the volume of the balloon. This

can be achieved with minimum input power by using a large number of turns of fine wire. A vehicle which is 1,000 feet in diameter and 500 feet high would consist of 400,000 cubes and carry a payload of 4,400 tons."

These points should be noted, said Hastings: One, the total payload is proportional to vehicle volume. Two, the electromagnetic torque and lift are also proportional to volume in this design. Three, navigation is achieved by manipulating the orientation of the coils, and, four, if superconducting wires become available, the batteries will not be required.

"When the Army and Navy start producing these machines and claiming them as their own, I want you to remember where you first saw it," Newman said.

Newman, who has had an ongoing battle with the U.S. Patent Office for the last eight years, said he is still taking his inventions to the public. "The public is not stupid," he said. "They will not let things like this be kept from humanity."

"I continuously push this technology forward for the benefit of humanity. I am pushing forward now vigorously attempting to get stockbrokers to come forward and give the public a chance to put forth the funds to get this technology into full production," Newman said.

FACT 40: It is obvious that Ronald Reagan has been advised to try to use his political position to obtain patents and pioneering patents for individuals claiming my work. At the same time Reagan is giving millions and possibly billions of dollars of taxpayer's money to forward the interests of energy concerns, but at a controlled rate of advance. He uses his "associate cronies" — Federal Judge Thomas Penfield Jackson and Donald Quigg — to slow down or stifle the Pioneering Patenting of my Technology. At the same time Reagan misuses this technology and increases the possibility of war through political foolishness. As has often occurred in history of our species, those least qualified to politically lead manipulate themselves into a position of leadership.

★ For further proof of the "Truth" of the facts presented in this Report, see page 22 for the extensive list of organizations, universities and corporations that have purchased my book.

THE MOVEMENT OF HIGH VOLTAGE DIELECTRICS

by
Doyle Buehler
Grade 11
Bishop Carroll
Calgary, Alberta

Experiments were performed to analyze the effects produced by a charged mass relative to the earth's Electric, Magnetic and Gravitational fields.

It has been noted in experiments that a simple parallel plate capacitor will exhibit a force when charged. The force acts in such a way as to physically move the entire capacitor in an upward direction, independent of external variables. (irrespective of polarity i.e.)

NEVER downwards! The main area of concern with regards to the experiments was that this phenomenon could not readily be explained in terms of current electromagnetic theory.

When the "home-made" capacitor was placed upon a sensitive scale and charged with 100 000 volts, the device proceeded to lose weight. It is to be assumed that a force was exerted downwards which moved the capacitor upwards, decreasing the scale reading. When the capacitor was charged the maximum force created was 4.0×10^{-2} Newtons, resulting in a net "loss" of approximately 2% of the weight of the entire capacitor.

The direction of the force apparently could not be attributed to Coulombs Law in relation to a negatively charged earth and a positive or negative charged capacitor. Coulombs Law is associated with the attractive or repulsive forces between opposite and like charges. The magnitude of the force created by the capacitor was also stronger than that theorized by a factor of 1000.

1/2 v² The outstanding feature of this "new" force was its relation to motion and energy. It was found that the upward force was in direct proportion to the amount of electrical energy that could be stored between the capacitor plates. A capacitor has the unique ability to "store" potential electrical energy. From this it is shown that upward motion is created by the use of potential energy and not kinetic energy. The application of this "law" may have immense possibilities in transportation and industry. Attributed to the energy law, the capacitor could theoretically work in a vacuum, such as outer space. Further extrapolation indicates that the capacitors could become self-elevating if sufficient energy was available.

There seems to be an extremely strange upward force produced by a charged capacitor. It apparently cannot be explained in terms of current electromagnetic laws. There is also a basic law concerning the movement, indicating that it isn't random. As to the exact nature of this force, it is still unknown, although important variables believed to be associated with the phenomenon have been eliminated. Further research will provide a definite base for a theory associated with the phenomenon.

-seems to be contrary to Brown?

-6
ELECTRODYNAMIC FORCES BETWEEN HYDROGEN ATOMS
COMPARED TO THEIR GRAVITATIONAL FORCES

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In the late 1960's, William Hooper, using some fundamental work on EM fields, proposed an electro-magnetic model to explain gravitation. The model required that the lines of magnetic flux associated with DC currents move at the current charge velocity. This model predicts a non-shieldable force due to motional electric fields that falls off inversely as distance squared. A previous paper discusses favorably Hooper's ideas about moving B fields; however, when the gravity force is calculated

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some disagreement results. The Bohr model of the hydrogen atom is used to find the force between two hydrogen atoms due to the motional electric fields created by the two electrons moving about their respective atoms. The Ampere force equation is used to predict the net force. This force has three terms. The first term is proportional to the square of the electron velocity and drops off as $1/R^3$. The second term is proportional to the velocity₄ and acceleration of the electron and drops off as $1/R^4$. The third term is proportional to the acceleration squared and drops off as $1/R^5$. The magnitude of the electrodynamic attraction is found to be within a couple orders of magnitude of the gravitational force between the two atoms, but the $1/R^2$ behavior characteristic of gravity has not been found. In light of recent theories of different materials experiencing different forces of gravity and the postulation of a fifth force or "hyper-charge", it is time to again pursue this basic work.

-G



MOTIONAL ELECTRIC FIELD THEORY OF FORCE
BETWEEN RELATIVE MOVING CHARGE

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A motional electric field approach for explaining the force between relative moving charge was first proposed by William Hooper. This approach comes directly from the assumption that the magnetic flux due to a current moves with the charge carriers that comprise the current. Motional electric fields are due to relative motion between magnetic flux and the reference frame in which the motional electric field is induced. Experimentally, motional electric fields differ from electrostatic fields in their immunity to shielding. Hooper's moving magnetic flux model is shown to yield the same results as special relativity for moderate charge velocities. Both predict a force between a stationary charge and a DC current flowing in a neutral metallic conductor even in configurations where the total magnetic field surrounding the conductor is zero. The difference between Hooper's theory and relativity is that they predict effects that have differing physical

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characteristics. Relativity predicts a small radial electrical field surrounding a conductor that is due to the increase in magnitude of the moving line of charge comprising the current and is electrostatic in nature. On the other hand, the motional electric field approach assumes that mass and charge are invariant and that this radial electric field is due to relative motion between the reference frame that current is measured in and the magnetic field surrounding the conductor. To date, Hooper is the only known investigator who has measured the radial electric field around a DC conductor and verified it having the properties of a motional electric field.

Force Field Shows Propulsion Promise

SAN DIEGO—A promising research approach to force field propulsion is reported by the Electronics Div., Ryan Aeronautical Co., San Diego, Calif.

The system is being studied for possible application in a series of manned watch satellites for detection and intercept of hostile ICBM's.

Martin N. Kaplan, a senior research engineer at Ryan Electronics, does not think present-day chemically fueled rockets have the duration required for an orbital mission. More exotic propulsion methods—nuclear, ion or plasma-type systems—can be ruled out because of weight or very low initial velocities, he says.

The need, according to Kaplan, is for a new approach to propulsion which would furnish very high speeds along with high initial velocities and long duration.

Ryan research in force fields is running along two paths that "look promising." One is termed "unified field theory," the other a "quantum mechanical theory of free space." The latter approach, derived from a quan-

tized model, has led to successful experiments, Kaplan says. Additional experiments are being set up.

A more comprehensive research program is planned which, if successful, will produce either "restricted" or "general" results. Either would be a revolution in propulsion.

Restricted results might be dubbed antigravitation. This would permit an acceleration force which could be directed only toward, or away from, a second body. A general result would permit a reaction against free space.

The general result is anticipated. But this would still require one major breakthrough—development of a means of converting potential energy (perhaps nuclear) to kinetic energy in the form of a rectilinear momentum without expelling mass. Such a breakthrough seems possible, Kaplan believes, when there is a better understanding of the mechanism by which matter is converted to energy.

• **100-g acceleration**—In practice, the thrust produced by engines of this type would be directly proportional to

their energy conversion rates. Neglecting energy losses, the vehicle's rate of change of momentum would be limited only by its peak energy conversion rate.

Kaplan believes these conversion rates could be safely employed to achieve space accelerations of 100 g or more. This is not so fantastic when compared to recent ultracentrifuge tests which have produced reactions against space of nearly 500,000 g. Once requirements for rectilinear reactions against space are discovered, 100-to-1000-g propulsion should be attainable.

A complete cycle of acceleration and deceleration at 100 g would take a vehicle from the earth's surface to a 1000-mile altitude in 80.8 sec., disregarding atmospheric drag. Although this extreme acceleration rate might be no problem for the vehicle structure, it would hardly create a friendly environment for its occupants or their instruments.

The same principle used for vehicle propulsion could be used to give the vehicle's occupants a g-controlled environment of a less disastrous character.

• **Putting it to work**—So where's the practical use if it all works? Kaplan proposes a system of six "stacked" counterorbiting fleets of 24 manned ICBM detection-pursuit vehicles per orbit and counterorbit. Vehicles in each orbit (12 in each direction) would be spaced about 2075 miles apart. Each fleet would patrol an area at least 2000 miles wide, extending completely around the world. Thus, 144 patrol vehicles would have to be in space at all times to give continuous early warning coverage of the entire earth's surface.

If, for some reason, orbiting vehicles could not be used, the vehicles might be placed in a hovering position, deployed to form a space blockade about a potentially hostile power.

The early warning system could double as repeater stations in closed communication links, separated by slightly over 2000 miles. Tandem vehicles of each orbit, or the vehicles of a hovering blockade, would be within line-of-sight of each other. Thus, they could relay information originating in any one station or from an earth station in contact with any one space point.

Method in Madness



BULGES AND BUCKLES with a purpose were built into this room at Republic Aviation's new Farmingdale, N.Y., Research and Development Center. Gibson-Girl build of walls makes almost completely echo-free environment for testing antennas.

ANTI-GRAVITY

The Science of Electro-Gravitics

DESCRIBED BY E. A. VAN ALLEN

HOW near are we to actual space travel? I doubt, personally if we will get to Mars with the limited power of our present rockets. Aviation and astronautics, however, stand on the threshold of a wonderful new concept—Electro-Gravitics.

New Type of Lift

Electro-gravitics is a vertical lift which is not accomplished by means of rocket thrust or propulsion by airscrews and airfoil and is not dependent on any type of atmosphere, air or a vacuum as would be found in outer space. All these methods seek to overcome gravity, but we approach a new era where man no longer seeks to leave our Earth by overcoming gravity, but instead by utilising it to his own advantage.

An anti-gravity machine is not impossible and many countries including Russia are at present investigating this new approach to aviation. Canada has its "Project Magnet" which is the production of an anti-gravity machine using the electro-gravitic principle. Many American aircraft manufacturers are spending millions of dollars on the use of gravity as applied to their industry. A number of universities are also going into the problem, which, incidentally, is not a new one. An actual flying model using this principle was made in England before the war.

Gravitational Irregularities

Before we go into how electro-gravitics work let us just consider gravitation for a few moments. The theory of gravitation as laid down by Newton in 1687 was accepted even with the discrepancies in the planetary motions which worried scientist and astronomer until this century, when Einstein proposed his new theory of equivalence and relativity. Even the new theories cannot account for gravity-defying irregularities which are noticed within the atom. Certain electrons are able to leap from one orbit to another, which results in light being emitted. (The gravitational theory applies to all bodies irrespective of their size and to account for this irregularity a new theory by Plank has been devised, called the quantum theory).

Also liquids do not conform, as is noticed by the flow of water against gravity in capillary tubes. Water will leap to a glass rod held just above it. Helium in liquid form acts even more strangely, as it will climb up the wall of its container and flow down the other

side. These have been classed as oddities in molecular arrangement, but above all they do illustrate that gravity is not altogether a force of attraction.

Relationship of Gravity and Electro-magnetism

Let us get back to Einstein who informs us that the attraction between atoms, molecules, planets and even sub-atomic particles is the same as the force of electro-magnetic attraction. Any body, if it be sub-atomic or as large as the greatest star has a magnetic field.

Einstein predicted and later proved that light when passing close to large bodies, such as our sun, is bent. The stream of electrons in a TV tube is also bent in the same fashion by a magnetic field. Here we see our first glimpse of the relation between electrified bodies and gravity.

Faraday, who found the relationship between magnetism and electricity, gave us the dynamo and electric motor. He also investigated the action of magnetism on light and his experiments proved successful. He then looked for the relationship between gravity and electro-magnetism, but failed to find it. This work was carried on by James Maxwell who showed the mathematical relationship between gravity and magnetism and found they obey the same laws. Einstein in his Unified Field Theory, put forward in 1950, also attempted to unify the laws of gravity and the laws of electro-magnetism. Up to the time of his death, he was perfecting this theory.

Under electro-magnetic forces we must include radio waves, heat, light, ultra violet rays, infra-red rays, X-rays and a host of others. Furthermore, the relationship with gravity must hold good for all of these.

Magnetic and Gravitational Fields

Let me now show how all this is connected with our anti-gravity machine. It is well known that if two fields of a magnet which are similar in character, that is to say two south or north poles, are brought face to face the magnets will be repelled. Similarly, two sub-nuclear particles which are the same, repel each other because they have the same electro-magnetic field.

If we can give our machine an electro-magnetic field which is similar in character to the gravitational field of our planet, we will leave this Earth at the possible speed of light. We would then travel along the lines of force of the gravitational field of our planet

October, 1961

NEW

as well as the lines of the fields of every Universe. Einstein maintains that these fields are ever present, have no end and spread in all directions, intertwine with each other in a complex fashion, but the strength decreases as the square of the distance from where they originate.

The problem is how do we switch our fields off? It has been found that there are particles being generated in the power accelerators which have the capacity to orbital motion of electrons and consequent magnetic field (as will be explained later) substance they are bombarding. The exact nature of these rays is not yet fully understood, but easily generated; furthermore, it does not change the orbital motion of electrons. It can be done in any physics laboratory.

Magnetism and Atomic Structure

Assuming that the reader has a knowledge of electron theory, I would briefly like to normal magnetism is brought about within the structure by changes rendered in the orbit of the electrons.

About our positive nucleus we have electrons spinning on different energy levels of which there are seven in number known as the K, L, M, N, O, P, and Q levels. The electrons not only spin on their own axis but also about their own axis. The electron is flowing in a definite orbit, it is a flow of electric current which would produce a current which would flow in a wire. Electrons do not rotate in coplanar orbits, but each atom would be a minute magnet with a north and south pole. It would arrange itself in neighbouring atoms so that they would serve in parallel directions which would form a magnetic substance with a powerful field. This is known as ferromagnetism.

There are, however, certain substances which have definite orbits of rotation, but the electrons of the same atom neutralise the magnetic field; because the electron spin on the axis for is in an opposite direction with respect to the electron on the same energy level.

The element iron for example has an atomic number of 26. The electrons in the K, L, M, and N shells are 2, 8, 14, and 2, respectively. The shell which is nearest the nucleus, contains the axial spin of which is considered to be neutralised, hence the magnetic fields are neutralised. In the L shell are eight electrons spinning in one direction, four in the opposite direction. In the M shell a difference of direction is thought to be spinning in and five in the other. In the N shell the spin is in opposing directions.

Why then does the unbalanced effect result in the alignment of all the atoms in a self magnetisation being produced? This is explained for atoms having unpaired electrons. This is due to the thermal disturbance of atoms and molecules at normal temperature. It has been found that in ferromagnetic substances a certain amount of alignment of atoms does not result in magnetism because the atoms in groups within the crystal lattices of are oriented in six different directions. Strongly magnetic, but is neutralised by groups which are oriented in five different directions.

as well as the lines of the fields of every body in the Universe. Einstein maintains that these fields which are ever present, have no end and spread out in all directions, intertwine with each other in the most complex fashion, but the strength decreases as the square of the distance from where they originated.

The problem is how do we switch our fields on and off? It has been found that there are sub-atomic particles being generated in the powerful nuclear accelerators which have the capacity to change the orbital motion of electrons and consequently the magnetic field (as will be explained later) of the substance they are bombarding. The exact nature of these rays is not yet fully understood, but they are easily generated; furthermore, it does not take much to change the orbital motion of electrons within the atom. It can be done in any physics laboratory.

Magnetism and Atomic Structure

Assuming that the reader has a knowledge of the electron theory, I would briefly like to explain how normal magnetism is brought about within the atomic structure by changes rendered in the orbital motion of the electrons.

About our positive nucleus we have the electrons spinning on different energy levels of which there are seven in number known as the K, L, M, N, O, P and Q levels. The electrons not only spin around the nucleus, but also about their own axis. Since our electron is flowing in a definite orbit, it is generating a flow of electric current which would produce a field, as current would which flowed in a wire. The electrons do not rotate in coplanar orbits, but if they did, each atom would be a minute magnet with a definite north and south pole. It would arrange itself with the neighbouring atoms so that they would align themselves in parallel directions which would give us a magnetic substance with a powerful field around it. This is known not to happen.

There are, however, certain substances which have definite orbits of rotation, but the electrons in the same atom neutralise the magnetic fields of the others because the electron spin on the axis for one electron is in an opposite direction with respect to the spin of an electron on the same energy level.

The element iron for example has an atomic number of 26. The electrons in the K, L, M and N shells are 2, 8, 14, and 2, respectively. The K level orbit, which is nearest the nucleus, contains two electrons the axial spin of which is considered to be in opposite directions, hence the magnetic fields of each are neutralised. In the L shell are eight electrons, four spinning in one direction, four spinning in the opposite direction. In the M shell a difference arises. Nine electrons are thought to be spinning in one direction and five in the other. In the N shell the two electrons spin in opposing directions.

Why then does the unbalanced effect in the M shell result in the alignment of all the atoms in the substance and self magnetisation being produced, as was explained for atoms having supposed co-planar orbits? This is due to the thermal disturbances of the atoms and molecules at normal temperatures. It has been found that in ferromagnetic substances there is a certain amount of alignment of atoms, but it does not result in magnetism because the atoms are aligned in groups within the crystal lattices of the metal and are oriented in six different directions. Each group is strongly magnetic, but is neutralised by the other groups which are oriented in five different directions.

This results in the substance being entirely non-magnetic.

The piece of iron is now placed in a magnetic field, such as a current carrying coil. The orientation in one particular direction is increased, which results in a change of direction of all the groups and brings about a powerful magnet. When the current is turned off the groups rearrange themselves and the metal is no longer a magnet. In steel there is a tendency for the groups to remain oriented in one direction after the current is switched off, hence forming a permanent magnet. Heat the metal and the magnetisation is lost due to the thermal agitation of the atoms. A constant change in the external field, as that of an alternating current, will also demagnetise the substance.

Diamagnetism

I have explained what happens in ferro and paramagnetic substances; there is a further class of substances which are said to be diamagnetic, all the non-conductors belong to this group as well as a few metals and alloys. A diamagnetic substance is one which has a small magnetic susceptibility and is weakly magnetised by a magnetic field in such a way that it moves towards the weakest part of the field and at right angles to the lines of force.

It will now be seen that this type of substance is repelled by the field in which it is placed, hence the field set up in the diamagnetic substance must be characteristically the same as the field in which it is placed. This type of magnetism is also due to orbital changes in axial moment of the electrons. This phenomenon occurs in all substances to some extent although the effects of ferromagnetism are greater and so more noticeable. The most diamagnetic substance known is bismuth.

Experiment in Diamagnetism

We can perform with suitable apparatus an experiment to demonstrate diamagnetism. This experiment is the basis on which an anti-gravity craft would operate and is also the starting point of the investigations on anti-gravity.

Our apparatus is simple. We need a powerful electro-magnet and a circular aluminium disc. This disc is placed on the solenoid and the current is switched on. A magnetic field is set up and aluminium being strongly diamagnetic is repelled with considerable force and speed. By shaping the disc so as to obtain an airfoil, amazing results can be obtained.

This then is the principle of our anti-gravity craft. We make our space-ship diamagnetic and travel along the lines of force of the planets. The machines would be constructed of such metals as aluminium and bismuth. This opens a new field for research to find alloys even more diamagnetic than bismuth.

We have seen that changes in orbital motion are easily brought about although it might be many years before we see machines utilising this principle in the sky. Perhaps the flying saucers are gravity ships manned by intelligences from space and using the gravitational fields of the universe as their tradewinds! We have yet to establish a really sound theory to relate gravity and electromagnetism. These problems are the ones that vex the scientists and we must patiently wait for our weightless anti-gravity machines of tomorrow. In spite of the new race to conquer space by rocket power, I am convinced that the exploration of our solar system will not come until we have our anti-gravity space ships.

THE SECRETS OF FLYING SAUCER PROPULSION

by
Noel Huntley

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If we can find out how to reverse the Force of Gravity and use its secret for man's propulsion wherever in the universe, or on earth, we may want to go, a phenomenal era of transportation and economic power will be opened to us.

THE AMERICAN MERCURY

Another Step Toward ANTI-GRAVITY

by Gaston Burrige

ONE OF the greatest energies on earth—even greater in the universe—is the Force of Gravitation. As a power potential, it probably outstrips atomic energy many times, with no deadly gamma rays as its aftermath.

The force of gravity is one of the great mysteries confronting mankind today. If we Americans can straighten out the question mark behind this, until now mysterious, force, and make it into an exclamation point, we need not care who will be first to reach the moon.

Sometime before 1923, Dr. Paul Alfred Biefeld, Professor of Physics and Astronomy at Denison University, Granville, Ohio, and

a former classmate of Albert Einstein, in Switzerland, noticed a phenomenon relative to the movement in an electrical condenser when charged. This observation was the actual birth of the so-called Biefeld-Brown Effect. The successful application of this Effect by Thomas Townsend Brown, during the last 30 years, appears now to have been accomplished;

our first anti-gravity device seems within our grasp.

Townsend Brown, however, has been something of a scientific outcast in the United States, because his views have been at odds with orthodox "opinion." His recent and essential research has been conducted in France, under French Government sponsorship. Realizing how riddled France is with Communists, it may well be that the United States no longer possesses all the known pieces of this scientific puzzle.

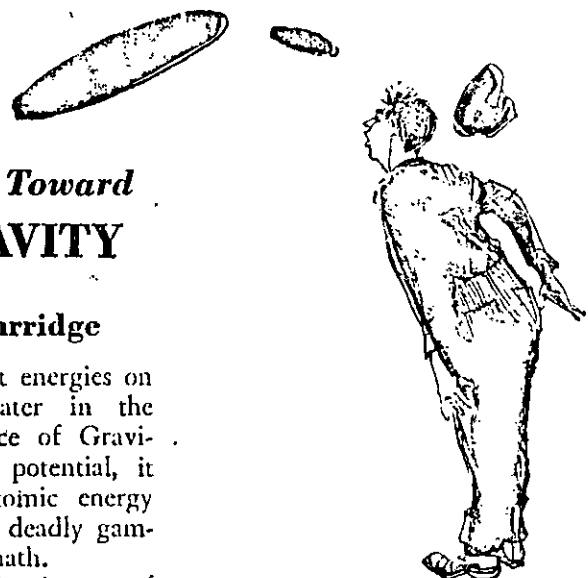
Townsend Brown was a student at Denison University in the early 1920's. A quiet, retiring young man then—as he is a reticent, middle-aged scholar, today—he has always had a great interest in the strange, unexplained mysteries of the electrical field, but his keenest interest attached itself early to the strangest of electrical phenomena, the movement which Dr. Biefeld had noted in electrical condensers when charged.

To capture the essentials of this phenomenon and to apply this movement toward an anti-gravity device became a passion in Brown's life, an unquenchable fire which has consumed his own considerable private fortune and most of his waking hours. If ever a man willed the application of an idea into reality it is Townsend Brown. He now has a new and very capable sponsor, one who long has been interested in the force of gravity.

AS FAR as science has yet determined, everything in our universe is motivated by only three energies: electricity, magnetism, and gravitation. From the greatest spiral star cluster down through the smallest atom known—hydrogen and its parts—only these three forces are active. Whether they are three distinct and separate energies, or separate phases of one force which we have not yet been able to recognize, appears to be a point of endless argument between physicists and metaphysicists. Nobody knows, yet.

Taken by itself, not one of the three forces amounts to much. A combination of electricity and magnetism, however, has boosted man to his present high standard of living. The potential of this combination is known to every high school physics student, and we are far from having researched its ultimate usefulness.

While the kinship of electricity and magnetism has been discovered and has become very applicable, a similar association between electricity and gravitation has remained most theoretical. American scientists do (unenthusiastically) acknowledge a *very loose coupling effect* between these two forces, but most of them deny that this coupling effect is at all applicable. This is where Mr. Brown and orthodox science part company, he believes the coupling force is strong and that it *is* applicable.



The connection between magnetism and gravitation remains still more nebulous. However, if there is a relationship between electricity and magnetism, and between electricity and gravitation, as Mr. Brown appears to have proven, then it would seem that an association between magnetism and gravitation also exists.

Mr. Brown maintains that, through the Biefeld-Brown Effect, a tie between electricity and gravitation has been established, brought about and held by the same means of an electrical condenser.

AN ELECTRICAL condenser is a device which will absorb electrical energy as an "elastic stress." In simplest form it can be two metal plates. Placed between them is a piece of material through which an electric current cannot ordinarily pass. This material, of such nature as to be able to absorb and hold electrical energy as an elastic stress, is known as a *dielectric*.

There are many different types of dielectrics—glass, hard and soft rubber, *air*, ceramics, paper, bakelite and various other plastics, each possessing its own special advantage in a particular use.

Some dielectrics are capable of absorbing huge quantities of electrical energy if fed them slowly at low potentials. Others, like lead-free glass, can be charged and

discharged hundreds of thousands of times a second at extremely high potentials!

How does an electrical condenser operate? Let us assume that we have a piece of sponge rubber four by six inches, two inches thick. It is the dielectric. We place this between our two hands (which are the condenser plates) and press—which is the charging potential or voltage. The plates (our hands) distribute the energy evenly over the rubber, forcing energy into a smaller space, where it remains until our arms cease to act; then the rubber springs back into its original shape. When the rubber is compressed it is similar to the condenser being charged. It is discharged when it "jumps" back into its original shape.

The Biefeld-Brown Effect proves that an electrical condenser will move toward its *positive pole*, and remain so positioned until discharged; that this movement will occur, regardless of which plate of the condenser is made the positive pole, or which side of the dielectric receives the positive charge.

The movement does *not* nullify the scientific "law" that every action carries within it an equal reaction. A reaction is present, but, as in the case of gravitation itself, the reaction is not obvious.

Discoid airfoils 24 inches in diameter, designed and built by Brown, have attained a speed of 17 feet per second, or about 11.8

miles per hour, in the Brown laboratories. These discs were a variation of the simple two-plate condenser, charged with 50 kilovolts, or about fifty thousand volts, of direct electrical current. Tethered to a mast, they spun in a circular course 20 feet in diameter. The continuous energy input to keep them flying is reported to have been only 50 watts, or that required to light a small bulb.

Another set of experimental discs three feet in diameter, also tethered to a central pole, ran a 50 foot diameter course. These discs were placed under 50 kilovolts also, and their speeds were reported to be so impressive as to be highly classified.

There is a slight hum emanating from the discs as they fly. In the dark they glow with a weird lavender light. They whirl in free flight, their power being supplied to them through wires from the central mast.

Many scientists and engineers have witnessed *these flying discs*. Their opinion has been, generally, that the motive force propelling them was one which is often called "electric wind." Few, if any, up to now, have believed the Brown discs were propelled by the new principle Biefeld or Brown had discovered.

Though Brown did not agree, there was little he could do until he went to France. There he "sailed" some of his discs in a

high vacuum with singular success. Not only did the discs fly more efficiently, but *as there was no air present there could be no "electric wind."*

THE FIRST empirical experiments which led to the present scientific flowering of the Biefeld-Brown Effect are these: suspend a simple, two-plate electrical condenser by a cord to allow as complete freedom of movement in all directions as possible—except of course, downward. When this condenser is charged with the proper amount and pressure of *direct* electrical current, it will swing from its vertical, uncharged position, to an angular one and remain there, quite evidently "defying gravity." The new position is always toward the positive pole of the condenser.

When this condenser is discharged, and the positive and negative wires reversed relative to their former positions—and the condenser recharged—the entire condenser swings in the *opposite* direction from the one assumed during the first charge. In both cases the condenser maintains its angular position as long as the condenser is kept charged. This angular movement is equal in both directions, regardless of which side of the dielectric is positively charged.

By proper electrical switching arrangements, quickly changing the sides of the dielectric upon

which the positive charge acts, it can be seen that a swinging motion, similar to that of a pendulum, can be attained. Further, it will be noted that if a proper series of condensers is attached to, and placed around, a movable axis—and the series of condensers is properly and rhythmically charged and discharged—a rotary motion of the axis will result.

Another experiment is by balancing a simple, two-plate condenser at one end of a beam. Place enough dead weight at the other end of the beam to balance and maintain the bar in a horizontal position, if suspended at its center. Now, charge the condenser. If the position of the positive pole faces "up" or toward the sky, the dead weight at the opposite end of the bar will drop downward. This indicates that the condenser has *lost* some of its "weight" on being charged.

If the positive pole of the condenser is placed on the bottom or facing downward, and charged, the condenser will appear to have gained weight and will *raise* the dead weight on the opposite end of the beam.

These two positions will be maintained as long as the condenser is charged, the original balance between them returning when the condenser is discharged! *This* is the application of the Biefeld-Brown Effect which is of the most importance antigravitationally.

It would appear, therefore, that a definite relationship between electricity and gravitation exists; that this relation is as potent as the one long established between electricity and magnetism, and that it possesses definite scientific future applicability, perhaps when applied practically in aeronautic and astronautic space flight.

At present, detailed answers cannot be presented—for several reasons. Not all the answers are yet known. Many of the known answers must await the granting of some 19 patent applications, now before the U. S. Patent Office.

It seems most probable, however, that a space vehicle using the Biefeld-Brown Effect will be discoid, the shape which appears most efficient. The discoid probably will be surrounded by a flat rim, about one quarter as wide as the diameter of the disc, making the vehicle look as if a ladder had been bent completely around it at the edge of the disc. The forward and rear portions of this "ladder rim" will be left open—with just the "ladder rungs" or spokes connecting the outer and inner rims. The side portions will be closed in solid.

The leading and trailing edges of this ladder rim will determine the direction of the disc's flight, the spokes or ladder rungs for the most part, acting as the condenser plates, the air between them being the dielectric. When charged, these

THE AMERICAN MERCURY

condensers produce a propulsive force, the front being positive, the rear, negative.

On reaching full charge, a condenser normally loses its propulsive force, but in this configuration the air between the spokes is also being charged; so, in principle, the charging force *can* be prolonged indefinitely, because it does not ever come to fulfillment while forward motion is desired. As the vehicle moves ahead, the charged air is left behind and the disc moves into new, uncharged air. Thus the propulsion is continuous. Thus, also, we *may* create, someday, actual space "flying saucers."

MAN's present use of power in jet and rocket propelled vehicles is called by some scientists a "sledge hammer" approach to high-speed, high-altitude flight. In the turbo-jet aircraft, man has increased vehicle thrust some 20 times. But he has achieved only a little more than twice the speed of the original vehicles. Is brute

rocket-force the best way to reach the stars? Scientists in France now doubt it.

The anti-gravity and propulsion properties of the Biefeld-Brown Effect should not be thought of as "perpetual motion." But as far as is known in our universe, nothing ever is totally gained or lost, just *used* while passing from a higher differential to a lower. Though in different forms, perhaps, *all that ever was, still is*—countless transitions to the contrary!

The force of gravitation is part of and a force in the universe. It exists and operates under scientific laws as rigid and precise as those governing electricity and magnetism. Why should we believe that gravitational force is beyond man's scientific control?

If man at last, successfully opens this puzzle box, all other power for propulsion may fade into horse-and-buggy history. We shall have passed through the jet and rocket age into America's Force of Gravity Era. We shall have achieved, at last, *ultimate* power.

Guidelines to Antigravity*

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(Received 12 September 1962)

This paper emphasizes certain little known aspects of Einstein's general theory of relativity. Although these features are of minor theoretical importance, their understanding and use can lead to the generation and control of gravitational forces. Three distinctly different non-Newtonian gravitational forces are described. The research areas which might lead to methods for the control of gravitation are pointed out and guidelines for initial investigation into these areas are given.

INTRODUCTION

EINSTEIN'S general theory of relativity provides a number of ways to generate non-Newtonian gravitational forces. Theoretically, all of these forces could be used to counteract the gravitational field of the earth, thus acting as a form of antigravity. The three outlined here were probably known by Einstein before he published his paper on the principle of general relativity in 1916.¹ They were first specifically derived by Thirring² in 1918, and since then have been contained in nearly every text on general relativity.³⁻⁵

Although non-Newtonian forces are well known to the theorists in general relativity, they are little known to those outside the field, and it was for this reason that it was felt that a simplified discussion of these unusual gravitational effects would be of interest to the nonspecialist.

The equations of general relativity not only predict the usual radial Newtonian gravitational force behavior of a stationary mass on a stationary test body, but they also predict that a moving mass can create forces on a test body which are similar to the usual centrifugal and Coriolis forces, although much smaller. In addition, when the general relativity field equations are linearized, they result in a set of dynamic gravitational

field relations similar to the Maxwell relations. Thus one can use intuitive pictures from electromagnetic theory to design theoretical models. Whether the effects predicted by the linearized theory really exist, will, of course, have to be checked by repeating the calculations with the nonlinearized field equations.

The essential point is that all of these unusual forces create accelerations which are independent of the mass of the test body and the forces are, therefore, indistinguishable from the usual Newtonian gravitational force.

NON-NEWTONIAN GRAVITATIONAL FORCES

Effect of Rotating Masses on Stationary Bodies

By using Einstein's general theory of relativity for a rotating system of masses, it can be shown that in addition to the usual Newtonian term, the gravitational scalar potential contains terms which arise from the rotation of the body. One of the shapes which has been rigorously investigated is the rotating massive ring.³

For a massive ring rotating in the x - y plane, the non-Newtonian acceleration on a stationary test body near the origin is approximately

$$\begin{aligned}\ddot{x} &= (MG\omega^2/2c^2R)x, \\ \ddot{y} &= (MG\omega^2/2c^2R)y,\end{aligned}\quad (1)$$

and

$$\ddot{z} = -(MG\omega^2/c^2R)z.$$

Where M and R are the mass and radius of the ring, ω is the angular velocity, $G=6.67 \times 10^{-11}$ $\text{m}^2/\text{kg}\cdot\text{sec}^2$ is the Newtonian gravitational constant, c is the speed of light, and x , y , z are the coordinates of the test body with respect to the origin of the rotating mass. From these equations it is evident that the rotating mass not only

FIG. 1. Non-Newtonian test body below

forces the test body into a circular orbit, it also causes the test body to move upward into the air.

Effect of

It has been shown that a rotating mass will exert a force on a stationary test body moving at a constant velocity. This force is proportional to the velocity of the rotating mass. Compared with the usual Lorentz force, it acts like the Lorentz force through a magnetic field.

One of the effects is the rotating acceleration on a test body inside the shell.

where v_z and the velocity of the mass and first term in type force or in the previous x and y components

* Robert L. Forward

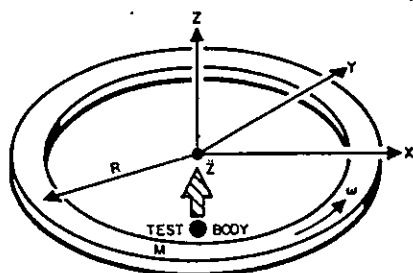


FIG. 1. Non-Newtonian component of acceleration on a test body below the center of a rotating massive ring.

forces the test body away from the axis in an imitation of centrifugal force, but also pulls it upward into the plane of rotation as shown in Fig. 1.

Effect of Rotating Masses on Moving Bodies

It has been pointed out that a rotating mass will exert a force similar to centrifugal force on a stationary test body; also, if the test body is moving at some constant velocity $\mathbf{v}$, it will experience an additional force which is proportional to the cross product of the angular velocity of the rotating mass and the linear velocity of the test body. The additional force can be compared with two others: mechanically, it acts like a very weak Coriolis force; electrically,^{4,6} it acts like the gravitational equivalent of the Lorentz force on a charged particle moving through a magnetic field.

One of the shapes which has been investigated is the rotating massive spherical shell. The acceleration on a test body moving with a velocity $\mathbf{v}$ inside the shell is approximately⁴

$$\begin{aligned}\ddot{x} &= (GM/3c^2R) [\frac{4}{3}\omega^2 x - 8\omega v_y], \\ \ddot{y} &= (GM/3c^2R) [\frac{4}{3}\omega^2 y + 8\omega v_x], \\ \text{and} \quad \ddot{z} &= -(8GM\omega^2 z/15c^3R),\end{aligned}\quad (2)$$

where v_x and v_y are the x and y components of the velocity of the test body and M and R are the mass and radius of the spherical shell. The first term in each expression is the centrifugal-type force on a stationary test body described in the previous section. The second term in the x and y components of the acceleration depends

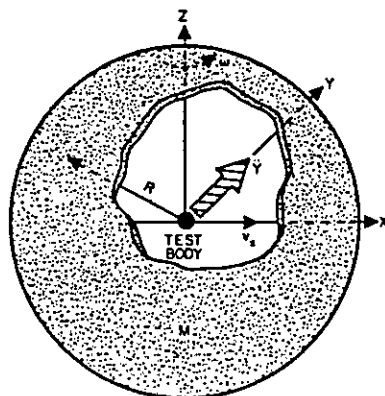


FIG. 2. Non-Newtonian component of acceleration on a test body with a velocity v_z at the center of a rotating massive sphere.

upon the velocity of the test body as shown in Fig. 2.

Effect of Accelerated Masses on Stationary Bodies

In employing Einstein's Theory to investigate the effect of a large accelerated mass on a small test body, it is found that the accelerated body drags the test body along with it. The exact equations are⁴

$$\begin{aligned}\ddot{\mathbf{r}} &= \nabla(GM/R) + (4GM/c^2R)\mathbf{a} \\ &\quad + (4GM/c^3)(\mathbf{a} \cdot \mathbf{R}/R^3)\mathbf{v}, \quad (3) \\ &= F_1/m + F_2/m + F_3/m,\end{aligned}$$

where M , $\mathbf{a}$, and $\mathbf{v}$ are the mass, acceleration, and velocity of the large body, respectively, and $\mathbf{R}$ is the distance from the small body to the large body. In addition to the usual Newtonian attraction, the test body experiences forces in the direction of the acceleration and the velocity of the large body as shown in Fig. 3.

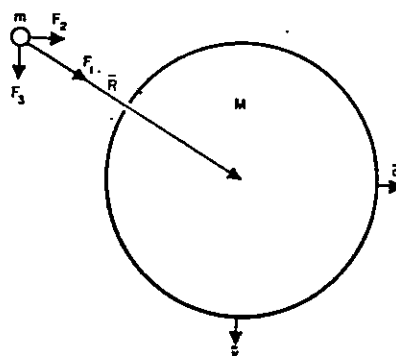


FIG. 3. Forces on a test body near a massive, accelerating, moving body.

⁴ Robert L. Forward, Proc. IRE 49, 892 (1961).

CREATING NON-NEWTONIAN GRAVITATIONAL FORCES

Devices Using Moving Masses

The equations in the previous section contain two common factors. One is the Newtonian gravitational field,

$$GM/r^2 \approx (4\pi/3)G\rho r, \quad (4)$$

and the other is the ratio of the characteristic system velocity to the velocity of light such as

$$v^2/c^2 \text{ or } ar/c^2. \quad (5)$$

In order to obtain measurable amounts of gravitational force, these quantities must be as large as possible. To obtain a high gravitational field, either a large mass or a high density is required. The greater the density, the less total mass necessary to achieve the same gravitational field.

To obtain high rotational velocities, we cannot use the mechanical strength of materials since this limits the obtainable equatorial velocities to approximately the speed of sound.⁷ Because of this, it will be necessary to use fields to hold the systems together under inertial stresses. One example would be the high gravitational fields obtainable with dense matter. However, any practical device which might be constructed would probably use electric or magnetic fields. By using electromagnetic forces to contain rotating systems, it would be possible for the masses to reach relativistic velocities; thus a comparatively small amount of matter, if dense enough and moving fast enough, could produce usable gravitational effects.

An example of a system held together by gravitational fields is a contact binary dwarf star system. The force equation describing the mutual rotation of the stars is

$$-(GM^2/(2r)^2) = -M\omega^2 r = -(Mv^2/r). \quad (6)$$

A particle coming near the system will experience not only a radial acceleration⁴

$$\ddot{x}_r = -GM \left[\frac{1}{(b-r)^2} + \frac{1}{(b+r)^2} \right] - \frac{4GMa}{c^2} \left[\frac{1}{b-r} - \frac{1}{b+r} \right], \quad (7)$$

⁷ J. W. Beams, Sci. Am. 204, No. 4, 134 (1961).

but also a tangential acceleration

$$\ddot{x}_t = \frac{4GMva}{c^2} \left[\frac{1}{b-r} + \frac{1}{b+r} \right] \approx \frac{1}{c^2 b} \left(\frac{GM}{r} \right)^{1/2}, \quad (8)$$

where b is the distance of the particle from the center of the system.

In applying these equations to a test object (such as a space vehicle) passing by this star system, it is evident that, in general, the radial acceleration will not introduce any net change in velocity, but that the tangential acceleration will transfer energy and momentum to the vehicle. For a neutron binary star, this acceleration can be greater than one million g's; although these accelerations seem very high, there are no stresses on the human body because the forces are gravitational. Such systems could be used to accelerate space vehicles to nearly the speed of light.

Devices Based on Analogies between Electromagnetic and Gravitational Fields

There are other types of devices for obtaining non-Newtonian gravitational forces which follow from the known electromagnetic analogies based on Einstein's general theory of relativity; however, these devices have not been analyzed using the complete field equations. For instance, two rotating gyroscopes should repel each other if oriented properly, and two pipes with massive liquid flowing through them should exhibit a pinch effect.

Although the rotating gyroscopes and flowing masses should exhibit non-Newtonian forces, the linearized theory does not predict an interaction with a stationary test body. A device which electromagnetic analogies predict might be able to create a non-Newtonian gravitational force that will accelerate a nonrotating, nonmoving body is one that contains accelerated masses whose mass flow is like the current flow in a wire wound torus.⁸ If we look at the electromagnetic model, a current I through the wire causes a magnetic field in the torus. If the current constantly increases, then the magnetic field also increases with time. This time-varying magnetic field then creates a dipole electric field. (Fig. 4.)

⁸ Robert L. Forward, Proc. IRE 44, 1402 (1961).

The value of the torus is

$E = \frac{1}{2} \epsilon_0 \mu_0 \frac{dI}{dt}$

where R is the radius of one of the tori, and t is the total time.

If we consider a massive liquid, the electromagnetic forces now become gravitational forces, and the quantities

where G is the gravitational constant, the total mass M in kg, and μ is the magnetic permeability.

It is evident that the quantities say that the flow in a gravitational field, however, in some manner allows for a magnetic field to be constructed and generated.

It has been measured that the strength of the general magnetic field must be controlled. The basic principle using magnetic fields is

The value of the electric field at the center of the torus is

$$E = -\dot{B} + (d/dt)(\mu N I r^2 / 4\pi R^2), \quad (9)$$

where R is radius of the torus, r is the radius of one of the loops of wire wound around it, and N is the total number of turns.

If we replace the wires with pipes carrying a massive liquid, then the known analogy between the electromagnetic and gravitational fields can now be used.⁶ All the electromagnetic quantities are transformed to their equivalent gravitational quantities to obtain

$$G = -\dot{K} = -\frac{d}{dt}\left(\frac{\eta N T r^2}{4\pi R^2}\right), \quad (10)$$

where G is the gravitational field generated by the total mass current NT , and $\eta = 3.73 \times 10^{-28}$ m/kg is the gravitational equivalent to the magnetic permeability.⁶ (See Fig. 5.)

It is important to notice in the above equation that the time derivative operates on the entire quantity in the brackets. One would normally say that all quantities are independent of time except the mass flow T . If the amount of mass flow increases with time, then $\dot{T} > 0$, and thus a gravitational field G can be generated. However, in electromagnetism, the permeability of some materials such as iron is nonlinear, which allows the construction of highly efficient electromagnetic field generators. A material with a highly nonlinear η would also be useful in the construction of efficient gravitational field generators.

RESEARCH AREAS

Dense Materials

It has been emphasized that in order to achieve measurable gravitational effects with moderate amounts of mass, dense matter is required. Thus, the study of degenerate matter could lead to the generation and control of gravitation. Methods must be found to manufacture, contain, and control matter with densities from 10^8 to 10^{15} g/cm³. The best starting point appears to be an investigation of the neutron-neutron interaction. By using extremely low temperatures and strong magnetic fields, one should be able to cool the thermal neutrons from a pile and concentrate

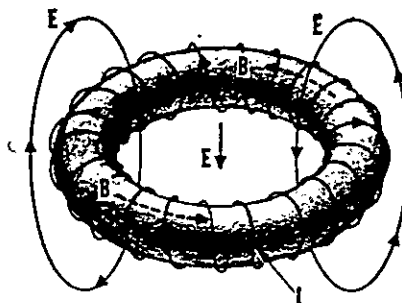


FIG. 4. Generator of a dipole electric field.

them into a small region through the interaction of the magnetic field with the magnetic moment of the neutron.⁹ The Fermi energy will limit the density to about 10^{-6} g/cm³, but the formation of tetra-neutrons (n^4) or the existence of a superconductive-type phase space condensation will create bosons which do not have this limitation. The results of a more comprehensive study of the properties of a cold neutron gas and the methods for containment will be given in a future paper.

Gravitational Properties of Matter

In studying analogies between electromagnetism and gravitation, it can be seen that one analogous quantity has not been investigated. This is the gravitational equivalent to the magnetic permeability.⁴ Electrical power distribution systems depend upon the anomalously large and nonlinear permeability of iron and other magnetic materials. Since all atoms have spin, all materials will have a gravitational permeability which is different from that of free space. Rough calculations show that this difference is very small, but experimental investigation may

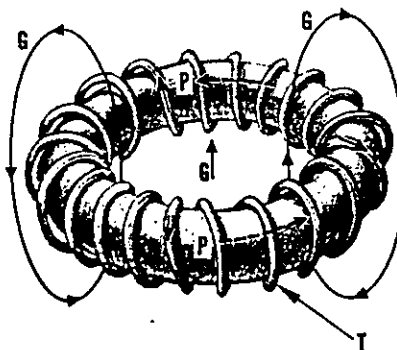


FIG. 5. Generator of a dipole gravitational field.

⁹ V. V. Vladimirovskii, Soviet Phys.—JETP 12, 740 (1961).

SPACE WARP I

by Charles R. Morton

1115 Summer Sun, Colby, Kansas 67701



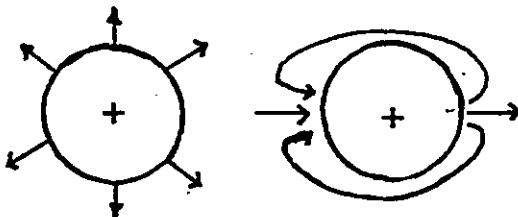
VAN DE GRAFF GENERATOR EFFECT

In the early 1960's, science and government were begging for a new method of space propulsion. This is an account of one man's attempt to solve the problem.

In the year 1966 I discovered that a Van de Graff generator gave off an explosive-like effect when it discharged. The explosion passed through plastic boxes but was reflected from metal. This effect and several other tests indicated an electro-magnetic nature but I was not able to find any physicist or engineer who could explain what the explosion was in scientific terms. One amateur told me it reminded him of a discovery called "the Brown-Biefield effect", but I was never able to find any more information on the subject.

VELOCITY ALTERS ELECTRIC FIELD

Then one day I set up an experiment in which the Van de Graff was set in motion and I made another discovery. **An electric field is altered by velocity.** Theoretically this is not possible except near the speed of light, yet this effect took place at a velocity of about 1 meter per second.



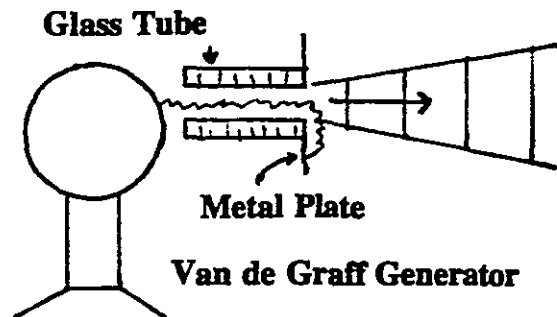
Charge at rest Charge in motion

The effect was to increase repulsion electric force ahead of the moving charge but to decrease attraction. Behind the moving charges, attraction increased but repulsion decreased. At right

angles, charges seemed to move in the opposite direction to motion. Once again science failed to give me any answer as to what was happening. But now I was pretty sure I knew what caused the explosion in the Van de Graff.

FORCE CONCENTRATION

In order to concentrate the force, I set up an experiment in which the spark fired at a grounded metal target.



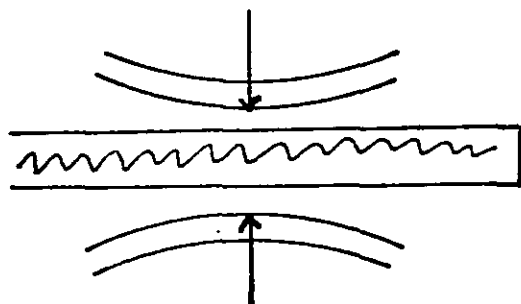
The spark fired through a glass tube toward a metal plate with a hole in it. From the tube came a beam of energy unlike anything I had ever heard of. **Plastic and glass ionized instantly, fluorescent tubes lit up, but unlike the Van de Graff explosion, this beam of force passed through metals.**

Some PhDs theorized I may have generated X-rays but the force was so powerful that it sent bits of paper flying. This would require millions of watts power if it were radiative pressure. Other PhDs said the field was high velocity ions, but the field showed no response to magnetic fields. One sure test of an ion beam is that it bends in a magnetic field.

BEAM ATTRACTS MATTER

As the years past, I developed better and better methods of producing the beam. Then one

day it happened—instead of repelling matter, the beam attracted matter. Even radiation pressure could not explain this phenomena. Furthermore, a lot of strange discoveries were made. When the spark went through the glass tube, the air collapsed around it.



This led me to theorize that perhaps the field of matter may collapse at relativistic velocities. So I tried several experiments in which uncharged bodies were accelerated at low velocity. Sure enough, accelerating balls and tops attract each other while decelerating bodies repel.

THE NEW PHYSICS

Eventually I, in desperation, turned to what is called the new physics. Here I began to find some answers.

Matter is not stable; it continuously breaks down into photon-like particles and fires them into other matter around it. Matter is also absorbing other particles which are fired into it. This creates standing wave fields around matter.

This field apparently collapses and expands with changes in velocity. It also creates a strange kind of radiative pressure called *relativistic waves*, which consist of standing waves in motion.

RELATIVISTIC WAVES

These waves may be electrical or gravitational in nature. To define the force they create one can say $h\nu/dt = \text{Force}$. But the energy necessary to create this kind of photon does not come from the input energy. It comes by virtue of the uncertainty principle which allows matter to decay into photon-like particles.

To define its effect on a gravity or electric field, one can say:

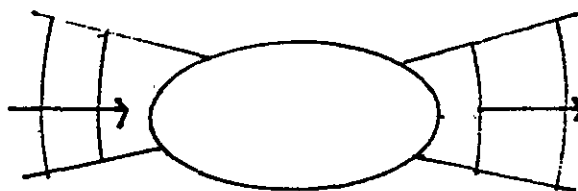
$$F = \text{static force plus or minus } h\nu/dt$$

Please keep in mind that relativistic waves only move when matter moves; they don't radiate in the same way light does. But they have a fantastically high frequency even at low velocity.

But relativistic waves do expand or collapse with changes in velocity and this rate of change can easily exceed the speed of light.

NEW SPACE DRIVE

Out of this research came a very strange kind of space drive. Out of one end of the ship comes a beam which continuously attracts matter, as long as it doesn't get too close. Out of the other end comes a beam which repels the entire universe of matter.



The craft model can develop several ounces of thrust yet no mass enters or leaves it. The effect is somewhat identical to a charge in motion, as though some charge continuously accelerated from one end of the craft to the other yet it repels and attracts both charges.

All tests indicate the beam is of a photon-like nature, yet it only requires a few watts of input power.

In theory, a full-scale craft could be built, which could develop several tons of thrust and exceed several times the speed of light. In many ways it would be identical to the space warp of science fiction.

It is my theory the craft model absorbs these photon-like particles at the attraction end of the craft and ejects them at the repulsion end. No test has ever indicated the presence of high velocity ions or any kind of mass, yet the beam feels like wind in spite of the fact it can blow through solid steel and glass. ###

THE PHILADELPHIA EXPERIMENT AND A MODEL ELDRIDGE OR GERMAN PROJECT V-7 DESIGN

The famous United States Navy tests in 1943 on the ship the Eldridge were called "The Philadelphia Experiment" in most of the literature and in the recent movie on the subject. The supposed purpose of these tests was radar invisibility via high powered EM generation. The supposed disasters were the invisibility and the location jump to Norfolk, Virginia and back to the Philadelphia Navy Yard.

The theory that cross-axial high powered EM radiation would deflect or absorb radar does not fit facts, then or now. Stealth designs and other minimum radar profile techniques are passive designs. The Navy got the idea for the experiment from the cross-axial EM experiments of Kowsky and Frost on crystals that levitated. The Eldridge was to be a giant metal crystal in effect, under the influence of this very powerful two axis (cross-axial) electromagnetic field. The Navy was experimenting with levitation (flying ships) which would be a tremendous weapons movement advantage in war. A war ship once levitated would have theoretically near zero inertia and could be easily propelled at hundreds of miles per hour to a new location for combat purposes. A twenty knot surface speed for a ship is nothing compared to a 500 knot flying speed. And just think about cargo movement to England and the Pacific fronts without losses to enemy submarines at those speeds.

The experiment on the Eldridge worked perfectly as far as the ship is concerned. It was levitated and flown from Philadelphia to Norfolk and back. The glow or cloud around the Eldridge was a smoke screen to cover up the disappearance, much like the stage props used by magicians. Who would look thousands of feet in the air for a missing war ship which moved off rapidly? So, of course it disappeared on surface-type radars and to human observers. Wouldn't you make up a cover story of "radar invisibility" tests to hide levitation weapons? The only problem was the effects on the exposed human crew. The human brain and body cannot tolerate massive, crude electromagnetic fields. Some of the crew went insane, some jumped overboard, and all suffered massively. There are no Eldridge survivors interviews anywhere in public information. Where is that crew? The Eldridge was sold or given to the Greek Navy after the war.

This levitation technology, which is very similar to the German V-7 levitating flight-plane project, went back to the bench for awhile. The theory was stolen from the Germans via the Canaris-Bulles spy group. The technology is simple enough to understand as a gravity levitation system. The simple electromagnetic coil levitator on pages 20 and 21 of this report explains it and can be modified to two axis to duplicate it. All that is necessary to do is make four coils that are placed vertically on a small square frame at the corners, just as the EM antennas were placed on the Eldridge or under V-7. This is how Kowsky and Frost levitated their crystal via a two axis EM feed system (but quartz crystals swell 8 times in volume under this EM input, while metals don't). Because of the use of electromagnets rather than Hertz EM, we cannot use iron to levitate the model, but aluminum will do just fine as the ships metal for the model Eldridge or V-7.

The easiest design for a model Eldridge or V-7 is to mount the four vertical electromagnets on the four corner edges of the aluminum plate itself. Place the coils vertically so that the electromagnets' Bloch Walls feed directly toward the center bottom of the plate from below. This means that the the coils are vertically less than one half above the horizontal aluminum plate and more than one half below the plate, like a table whose legs stick up at an angle. Adjust the height and tilt of the four coils with respect to the plate for maximum levitation. This design aids Earth's gravity push to cause the levitation effect. This model Eldridge or V-7 is best tested out of doors and should be handled with extreme caution. Be sure that the A.C. circuit breakers can take the current load unless D.C. is used for the electromagnets. Allow plenty of wire lead length to the coils to permit lift off of the model Eldridge or V-7. Only turn it on for a brief moment as it will lift beyond wire length and crash. Add appropriate current controls for a hovering mode of controlled levitation and only operate it for 5 to 10 seconds at most, due to coil heating problems.

If the type of coils called for on pages 20 and 21 are used, a total levitating Eldridge or V-7 platform weight of 20 to 30 pounds at least should work. The total weight is the plate and the four coils plus wire drag all added together. Once the design idea

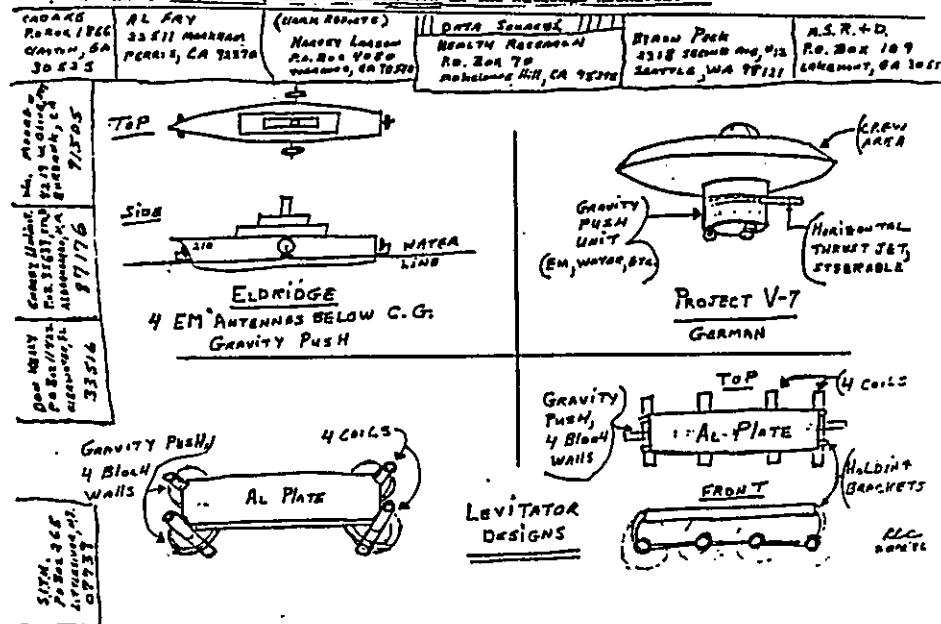
(31)

DR. R. L. CLARK :

is understood much better levitating platforms can be built. For example, Mu metal coil cores that are heat radiating finned inside would improve the basic design. An aluminum foil "saucer" shell and steerable horizontal thrust make a complete saucer model.

All that this design has done is to take the levitator on pages 20 and 21 and attach it to the levitated plate. A considerable part of the four coils' gravity (Bloch Wall) energy is directed from below into the plate but it cannot escape the levitating force in this design. In fact, you could design the four coils as an undercarriage attached lift unit, exactly expanding the original levitator on pages 20 and 21.

Attached are Schauburger notes and pictures. The full information on Project V-7 and the Philadelphia Experiment are hidden in the National Archives.



find materials with anomalously large or non-linear properties that can be used to enhance time-varying gravitational fields. Also, since the magnetic moment and the inertial moment are combined in an atom, it may be possible to use this property to convert time-varying electromagnetic fields into time-varying gravitational fields. At present, the only way to search for such materials is to intersperse wedges of material between gravitational wave generators and detectors, such as those described by J. Weber,¹⁰ and look for a change in amplitude or direction of propagation. The first efforts in this direction¹¹

have been carried out by the Russian workers Braginsky, Rudenko, and Rukman with negative results.

It is obvious that research in the field of gravitation will be very difficult since even the most optimistic calculations indicate that very large devices will be required to create usable gravitational forces. Antigravity, like space travel, will probably have no direct effect on the daily life of the average person. Future progress in the control of gravitation, like all modern sciences, will require special projects involving large sums of money, men, and energy.

¹⁰ J. Weber, *Phys. Rev.* 117, 306 (1960).

¹¹ V. B. Braginsky, V. N. Rudenko, and G. I. Rukman,

Zhur. Eksp. i Theoret. Fiz. 43, 51 (1962) (in Russian with English abstract).

PRODUCT ENGG. DEC 30 1957

Electrogravitics: Science or daydream?

A few weeks from now, at a special session of the Institute of the Aeronautical Sciences (New York City, Jan 27-31), a group of dedicated men will discuss what some people label pure science-fiction, but others believe is an attainable goal. The subject: electrogravitics—the science of controlling gravity.

Gravity is certainly a well-known phenomenon. Every schoolboy can "explain" it. Yet it remains a baffling mystery. For instance, as David B. Witty noted in his award-winning essay for the Gravity Research Foundation, gravitational screening is crucial in all theories of gravitation. If there were no way to block out the force of gravity or cut it off, "gravitation would be almost unique among physical phenomena." Yet there is no evidence of gravitational screening in our normal, everyday experience.

Theories proposed to explain this situation, include a complete reversal of conventional ideas. For example, E. M. Gluhareff, Pres. of Gluhareff Helicopters, suggests much progress might come if gravity were considered as "push" rather than "pull"—with all matter being pushed toward the

center of the earth by a sort of "electronic rain" from outer space.

Charles Tilgner, Jr., chief aeronautical engineer at Grumman and chairman of the forthcoming IAS session, suggests at least four ways that gravity might be controlled. He asks if the earth itself ever acts as a gravitational shield—a question also raised by Gluhareff's suggestions that gravity be considered as an external force.

Right now, as might be expected, the most promising approach to understanding and control of gravity seems to be via nucleonics. Such recently-discovered atomic fragments as hyperons and k-particles (Dec. 2, p. 16) appear to interact with nuclear matter in ways not explained by present theories. Scientists are now suggesting that these interactions may explain—or be explained by—gravity. It is even possible that gravitational energy may prove to be transformable into particles of this type.

Obviously, control of gravity is not yet just around the corner. Perhaps British aeronautical engineer A. V. Cleaver is right in insisting that if any anti-gravity device is to be developed the first thing needed is a new principle in fundamental physics—not just a new invention or application of known principles.

Nevertheless, the Air Force is encouraging research in electrogravitics,

and many companies and individuals are working on the problem. It could be that one of them will confound the experts.

How You Can Explore HIGHER DIMENSIONS OF SPACE AND TIME

T. B. PAWLICKI

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Dec. 15, 1942.

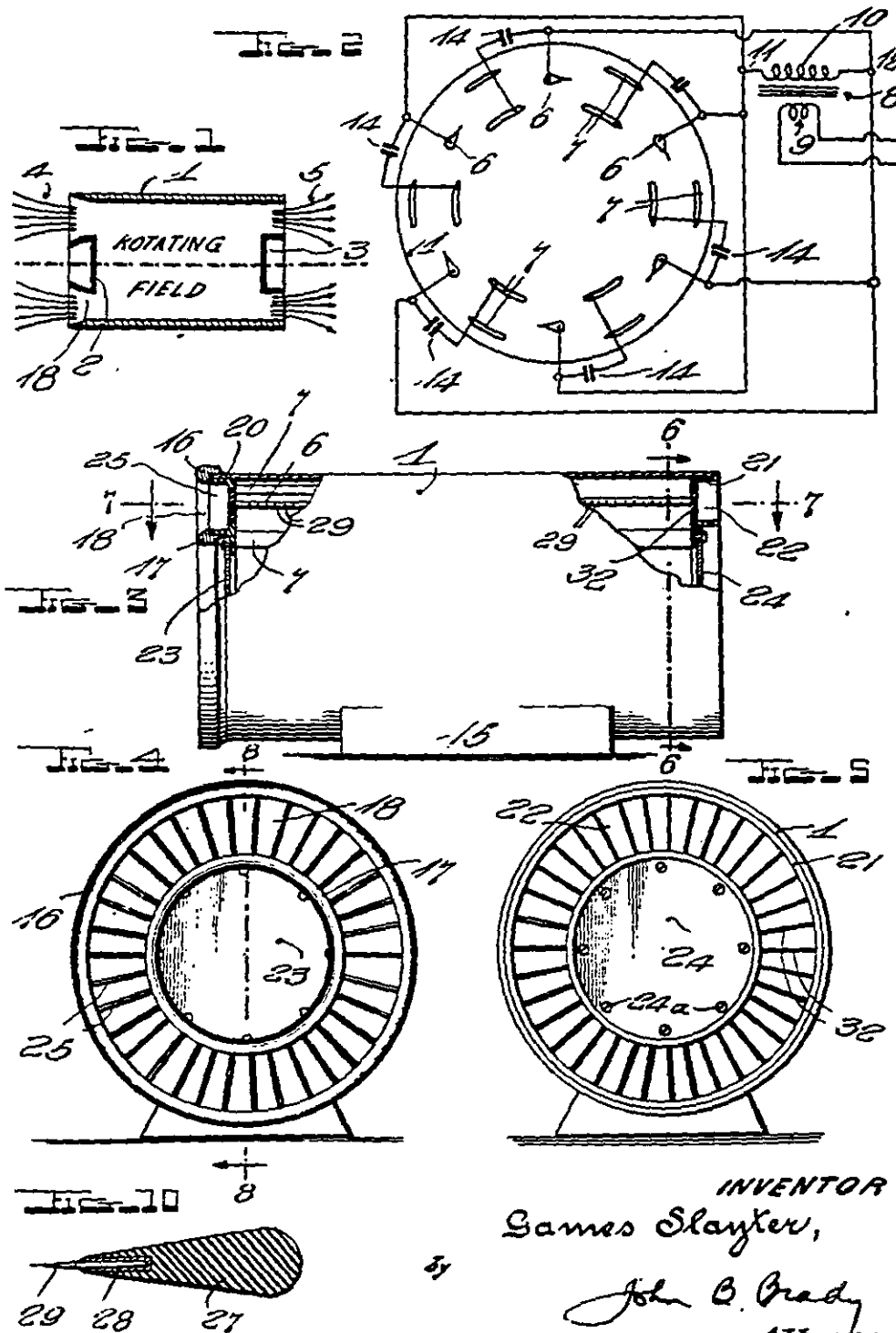
G. SLAYTER

2,305,500

APPARATUS FOR ELECTRICALLY GENERATING PRESSURES

Filed Jan. 22, 1940

3 Sheets-Sheet 1



Giant spring should detect gravity waves

NEW SCIENTIST
29 AUG '85

Marcus Chown

PUTTING a 25-kilometre long spring in orbit would, at first sight, seem an odd thing to do. But not to two physicists, Vladimir Braginsky of Moscow State University and Kip Thorne of the California Institute of Technology (Caltech). They champion the view that such a spring, or rather a cable with a spring at its centre, would make an inexpensive and sensitive detector of gravitational waves. They present the details of the detector, which they call a "skyhook" in *Nature* this month (vol 316, p 610).

The theory of general relativity predicts that gravitational waves are emitted by any mass which is accelerated, but as yet no one has detected them. They are difficult to detect because gravitational forces are very weak. For instance, the electric force between two electrons is 10^{42} times stronger than gravitational force. Only astronomical bodies possess enough mass to produce appreciable amounts of gravitational "radiation". Astrophysicists estimate that the strongest source in the Universe would deliver less energy to a detector than a lighted match placed at a distance of a million miles.

The skyhook detector proposed by Braginsky and Thorne would consist of two 20-kilogram masses, one on each end of the 25-kilometre cable. The cable, just 0.6

millimetres in diameter, would weigh about 20 kilograms, and the spring would be at its centre. In orbit, the cable would extend radially out from the Earth. The difference in the amount of gravitation pull from the Earth on the two masses would keep the cable taut.

Gravitational waves can be likened to ripples on the surface of a pond. Only they are ripples in 4-dimensional space-time, propagating at the speed of light. For waves on a pond the source might be a dropped stone. The source of gravitational waves, on the other hand, might be a supernova explosion, a disturbance at the heart of a quasar or simply two massive stars orbiting each other. Just as two ping-pong balls on the surface of a

pond would oscillate to and fro when passed by ripples, so the masses at the ends of the skyhook would jiggle about when struck by gravitational waves.

The skyhook would respond to gravitational waves in the 10 to 100 millihertz band. These waves are inaccessible to Earth-based detectors, because they cannot be distinguished from the constant background of Earth tremors at similar frequencies.

If a gravitational wave passed the sky-

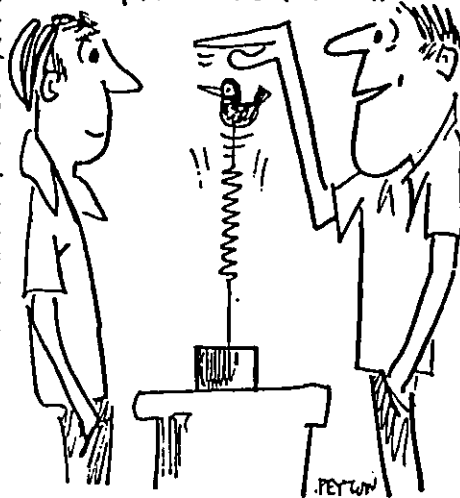
hook, it would make the masses at the ends of the cable vibrate back and forth. The force created by the oscillations of the masses would be transmitted along the cable to the spring, making it expand and contract. The force of the spring's expansion would then be detected by a piezoelectric crystal. Such crystals produce a potential difference across opposite faces when subjected to mechanical stress. In other words they produce a measurable, but transient electric current. They are capable of detecting stresses caused by movements as small as 10^{-15} centimetres—that is about a ten millionth of an atomic radius.

But gravitational waves are not the only phenomenon which could make the skyhook oscillate. Braginsky and Thorne recognise several potential sources of confusing "noise". All must be kept to a minimum if gravitational waves are to be confidently detected.

One source of noise might be any sudden change in the heat radiation received from the Earth and Sun. This could change the temperature of the skyhook, causing it to expand and contract in length. Painting the cable white to reflect light would minimise the problem.

Another noise source is tiny irregularities in the Earth's gravitational field. These occur over dense portions of the Earth's crust and could make the cable oscillate by tugging it unevenly. The only way to avoid this is to fly the skyhook in a high orbit at least a 1000 kilometres above the Earth. Also, the skyhook must not be made of a conducting material, because the Earth's magnetic field could cause currents to flow. These would interact with anomalies in the Earth's electric field, and would make the cable vibrate. □

THAT'S THE PRINCIPLE

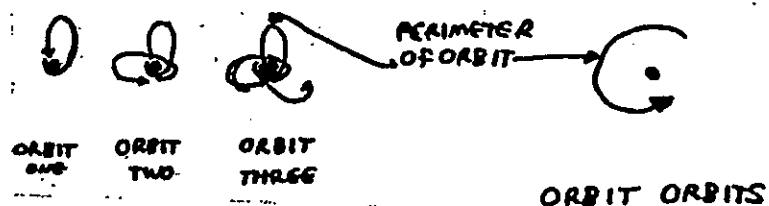


INTRODUCTION

This is an overview report on the Tesla Space Drive supposedly invented by Nikola Tesla and brought to light by one of his pupils who got it out of Tesla's files.

THEORY

Tesla's space drive is similar to the action experienced by an electron as it 'orbits' the nucleus of an atom. The electron orbits the nucleus in an elliptical orbit. This orbit is not static. That is, the orbit orbits about the nucleus also.



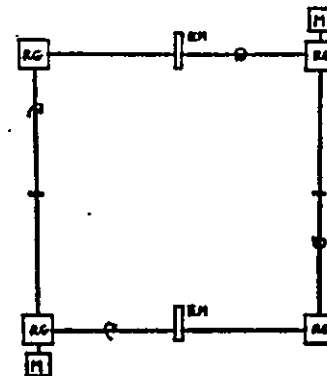
Theory has it that the nucleus has a hidden capacity to control its spatial displacement. This means that the nucleus reacts to compensate for any action that would interfere with its stable existence. It is like a gyro.

By the use of mechanics, Tesla's space drive creates the same effect and maintains this 'action' to cause a constant unidirectional force. Thus creating a means of propulsion and a space drive.

THE DRIVE

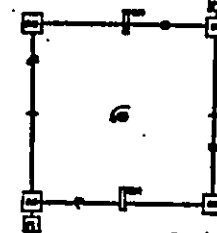
Tesla's drive is basically a one-body system. This means that for the whole rotating body(drive) each point within that body, each which has a different radius from the body's center, has a different orbit. In this one-body system each orbit has the same radius, but the center points of the orbits aren't the same. Mechanically we have an ungainly looking apparatus.

In this system the whole drive rotates counterclockwise with a superimposed counterclockwise gyration. This gyration force causes the drive to accelerate at a right angle to itself(straight out of this page).



OPERATION

The eccentric masses(EM)are accelerated about their axis as shown by motors(M) through right angle gears(RG). This causes a counterclockwise acceleration of the center of mass of the four eccentric masses as a whole about the x axis(coming out of the paper). This also produces a gyration reaction upon the rest of the drive. Remember that in this one-body system each orbit(of the eccentric masses) has the same radius, but the center points of the orbits aren't the same. This causes the center of gyration to lie between the center of masses of the drive as a whole and the center of mass of the eccentric masses. We have now created the same situation that exists with the orbiting electron and the nucleus. But unlike the 'free moving'



electron that the nucleus can 'use' to stabilize itself, our mechanical electron & nucleus can not. Therefore a force is generated. This force is along the x axis and comes straight out of the paper. This force can be thought of as a field since there is no mechanical reaction.

CONCLUSION

As we see this drive it involves too many moving parts. We are after a solid state drive. If you used an 'aligned nuclei' type of technology you could then construct a solid state drive.

RESEARCHER: L. Arkliniski EDITOR: Steve Clark

THE NIPD II SYSTEM

FOR FURTHER INFO:
CETAR, BOX 2252
COVER D'ALENG ID
83814

This short theoretical summary and the attached plans should enable you to construct a small flying model of the exciting NIPD II flight system.

↑
OUT-OF-BUSINESS
~~SECRET~~

It is our hope that you will enjoy building this model and that you will demonstrate it for your friend. We are counting on you to help us make "Antigravity" a practical reality.

Theoretical Data

Classical physics and General Relativity both assume that the gravitational constant (G) is unchangeable. This is the central concept of the equivalence principle which states that inertial mass and gravitational mass are the same thing and the weight of a body is directly proportional to its mass. However, while weight and mass are proportional to one another under normal conditions, it should be noted that they are different entities. Weight is the vertical force of gravity, mass is an inertial property. The mass of an object referred to in the law of gravitation is called gravitational mass, in contrast to the inertial mass. Einstein used this seeming equality of the inertial and gravitational mass as a basis for the General Theory of Relativity.

Dr. Erwin Saxl, a one-time student of Einstein, has experimentally proven that the assumption of a immutable gravitational constant is incorrect and obsolete. These findings were published in the July 11, 1964 issue of the highly-respected British Scientific Magazine NATURE.

By introducing the concept of dynamic interaction between electricity and gravity, we come upon a new and breathtaking view, which could explain a great deal. In essence, what Dr. Saxl did is this: Instead of testing the gravitational constant in a motionless system, he set about to study it under dynamic conditions. He built a system utilizing a huge ceramic disc rotating freely from a solid suspension specifically designed and constructed to prevent vibration or any outside interference. Then he found a way of regenerating the rotation of this "pendulum" electronically under exceedingly precise control. Finally, using a light beam and a photo electric cell, he succeeded in measuring the time it took the pendulum to swing over a certain arc with accuracy of one part in 10 million!

Saxl, found that if he charged the pendulum positively the period of swing for this torsion pendulum was increased. If the value of G was a constant no variation in period would have been noted.

Extensive additional experiments with his instruments indicated, that there is a definite relationship between gravity and electricity. A relationship which has not formerly been confirmed experimentally.

In 1923, Professor Paul Biefeld of Denisen University in Granville, Ohio suggested several experiments to T. Townsend Brown. These experiments led Brown to develop the Biefeld-Brown Effect and ultimately to obtain six patents on electrogravitational devices.

Brown's first experiment involved condensers charged with electricity.

III.

The startling revelation was that when placed in free suspension with the poles horizontal, the condensers, when charged, exhibited a forward thrust toward the positive pole. A reversal of polarity caused a reversal of polarity apparently caused by a reversal of the direction of thrust.

Further development of the implications this phenomenon illustrated a true "Antigravity" effect. When balanced on a balance beam, and then charged, the condenser moves. If the positive pole is up, the condenser moves up (i.e., becomes lighter); If the positive pole is pointed down it moves down (i.e., becomes heavier).

These two simple experiments demonstrate what is now known as the Biefeld-Brown Effect. To the best of our knowledge, it is the first method known of affecting a gravitational field by electrical means.

By experiment, Brown has developed an electrostatic propulsion method which was proven on a scale model going around a stationary pole. This model was tested in a hard vacuum at Bahnson Laboratories in Winston Salem, N.C. (Pat. No. ^{1,240}2,949,550).

One highly successful design based on the Biefeld-Brown Effect is shown in the book Antigravity Propulsion Devices by Bernard C. Ebershaw. (Patent ³#3,187,206). It is a rather sad commentary that the vast majority of aerospace engineers have never heard of Brown's work.

Except for a 1956 article in the prestigious international aeronautical journal Interavia and another article in the Journal of the British Interplanetary Society there has been virtually no widely published accounts of Brown's work.

The Biefeld-Brown Effect is the basis upon which the NIPD II system was developed. The Biefeld-Brown Effect is only effective at very high potentials (50KV or more). Such potentials are subject to dielectric decay in the atmosphere due to ionization. Therefore it was necessary to design the NIPD II system so that it could operate efficiently in the atmosphere near the earth.

This was achieved by combining the Biefeld-Brown effect and the well known concept of ion-wind into one design. Thereby the principle lifting force for launch is achieved using an ionic propulsion effect. This effect diminishes as the unit rises into the more rarified parts of the atmosphere. However, the thinner air permits an increase in potential leading to a larger Biefeld-Brown thrust component.

In the vacuum of space the Biefeld-Brown thrust component becomes the sole means of propulsion. The attached plans are for a small flying model of the NIPD II aircraft. No attempt is made here to incorporate manuevering and guidance system.

The NIPD II unit consists of a upper "antenna" grid and a lower "lifting" grid. The upper grid is connected to the positive terminal of the power supply and the lower grid is connected to the negative terminal. As defined by the Biefeld-Brown Effect, the NIPD II vehicle will move upward toward the positively charged grid.

The construction of the NIPD II flying model and its accompanying power supply is relatively simple. The total cost of materials should not exceed \$100.00.

Performance

No lifting force will be apparent when the high voltage potential is below 5000 Volts. Between 12,000 and 15,000 volts the vehicle should lift-off from the ground and float freely in the air. By increasing the potential between the two grids, the thrust is increased. At low altitudes this potential should be limited to 60 KV because of losses associated with corona discharges.

In the vacuum of space a voltage in excess of 250 KV will enable you to make maximum use of the Biefeld-Brown Effect.

The power supply plans provided in this information kit are for a power supply with 20 KV maximum potential. This was chosen for the low-cost demonstration version capable of illustrating the principle. You may design your own power supply to provide a higher potential. One low-cost alternative is a large Wimhurst-type electrostatic generator.

POWER SUPPLY RECOMMENDED FOR NIPD II

This circuit was selected because it is a simple, low-cost solution to the problem of generating the 20,000 volt potential necessary for flying the NIPD II prototype.

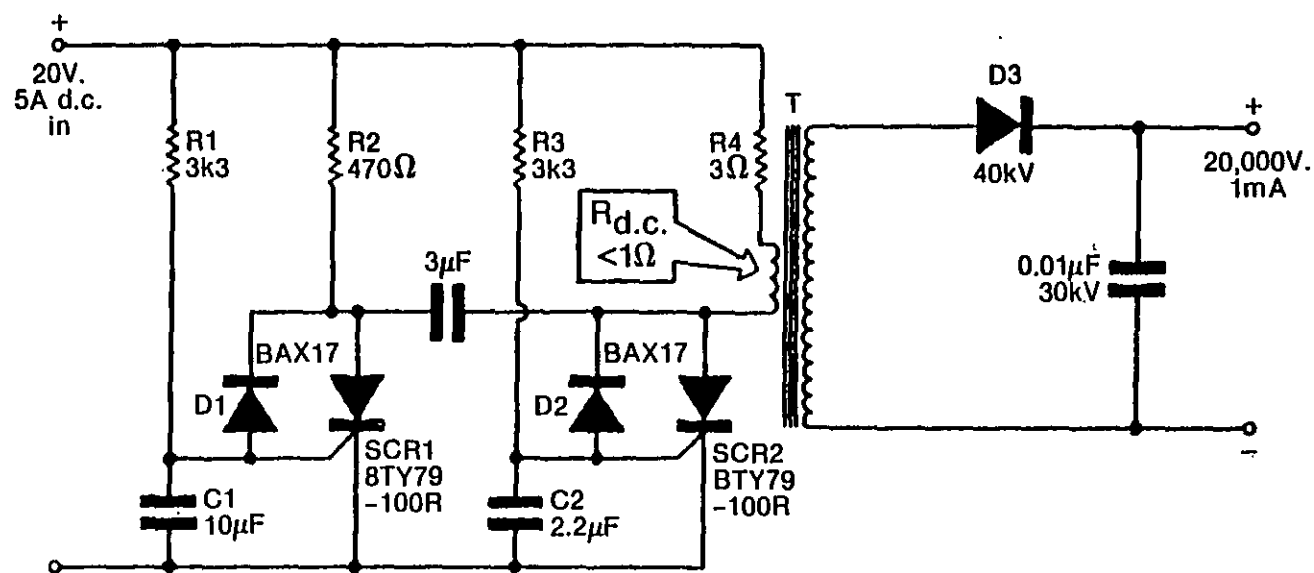
Two SCR's are used in an astable multivibrator with the primary of a high-voltage secondary transformer in the anode of SCR2. The rapid turn-off times of this SCR in switch-over provides the inductive kick to the primary of the transformer T. With the 200:1 step-up ratio, the secondary voltage rises to 20,000V.

The low-frequency multivibrator is assymmetric, owing to the approximately 5:1 ratio of the time constants of C1R1 and C2R3. This provides less than 10% duty cycle for SCR2.

The diodes D1 and D2 serve to discharge C1 and C2 during the "on" time of each SCR.

A standard automobile ignition coil can be modified to the 200:1 ratio required for the high voltage transformer.

The output is conventionally half-wave rectified by a 40KV diode and a 30 KV, 0.01 micro-Farad capacitor. The phasing of the half-wave rectifier is important as the circuit is of the ringing-choke type, where the output rectifier conducts-on the switched-off condition of the primary driving SCR.



D1, D2 = 200V, 200mA SCR1, SCR2 = 100V, 6A T = 200:1 step-up

20,000V, 1mA d.c. output from 20V d.c. input using s.c.r.s.

DIAMAGNETIC REPUSSION LEVITATOR

CONSTRUCTION

ORIGINALLY FROM CETAR

RICHARD LEFORS CLARK, Ph.D.
4015 CROWN POINT DRIVE, P-3
SAN DIEGO CA 92109

NOTE: Please read through this whole book carefully before you begin to build.

Here's what you'll need and what to do with it.

PARTS

- 3 inches of a 2 inch diameter pipe (IRON)
- 1 roll of electrical tape
- 185 feet of 15 gauge magnet wire. Copper, shellaced wire. (14 or 16 GAUGE OK.)
Don't use insulated wire.
- 1 switch, AC, 120 volt, 20+ ampere rating. Push ON, normally open.
- 1 plug, AC outlet type.
- 5 inch square aluminum plate. To insulate pipe from wood base.
- 10 inch square metal plate. This is the plate to be levitated.
Use aluminum, silver, or copper, some non-ferrous metal.
- 12 inch square plywood for base and parts mounting.
- 1 guideway. You will need something to guide your levitating metal plate. Cardboard or wood box. Plastic pipe. Other non-ferrous tubings or guideways. Or you can just tape your plate to the iron pipe as directed later.

$\frac{1}{4}$ inch
4 lb. lead = $\frac{1}{4}$ inch

DIRECTIONS

$$H \approx 50 K \quad K \approx 0.11$$

$$V = 1600 \text{ cm}^3$$

Wrap your 2 inch iron pipe with a couple layers of electrical tape to insulate it from the magnet wire.

Now wrap this with 350 turns of your magnet wire, leaving about a foot of each end for leads.

Connect the two ends of the magnet wire to the switch and then the switch to the AC plug. Don't plug it in yet.

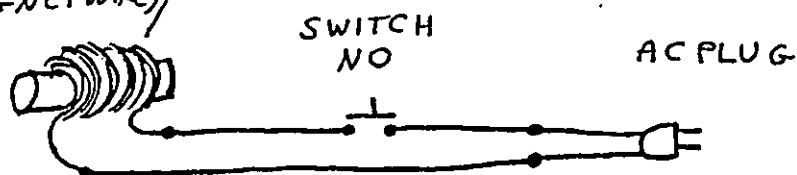
DIAMAGNETIC FORCE (IN DYNES)

$$F = K V H (dH/dx)$$

(CLASSICAL PHYSICS)

VERTICALLY dH/dx IS THE
(3) SQUARE OF THE DISTANCE
ABOVE THE POLE, FIELD
STRENGTH LOSS.

LEVITY GENERATOR
(IRON PIPE WOUND
WITH MAGNET WIRE)

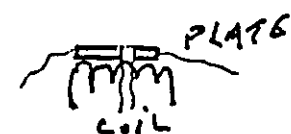


*NOT TO SCALE

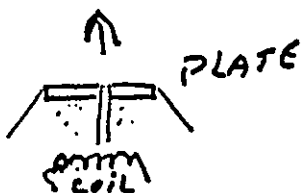
PARAMAGNETIC LEVITATOR SCHEMATIC

Mount the parts, except the AC plug of course, on the plywood base. Place the levity generator (wound iron pipe) on a plate of aluminum. Drill some holes through the aluminum and the plywood to mount the plate to the base. Put some washers between the plate and the base so they don't touch. The aluminum plate will get hot from the pipe levity generator and we don't want the plate touching the wood because it might burn it. If you are going to use a guideway make adjustments for its placement. Remember the levity generator gets hot so keep the guideway insulated from or far enough away from the generator. It would be much easier for demonstration purposes to just tape your non-ferrous metal plate to the levity generator and leave enough slack so it can rise up and levitate a few inches high.

REPELS

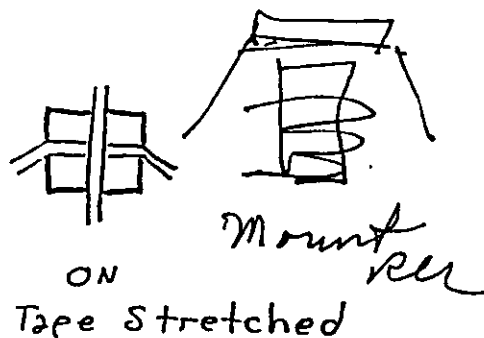


OFF
Tape Slacked



ON
Tape Stretched

SIDE VIEW



TOP VIEW

OPERATION

Turn the switch off, if you need to, and place the AC plug into a 120 volt outlet (110 volt is ok too). Your levity generator coil will heat up so pay attention! Turn the switch on for only 10 seconds at the most! Then turn it off and allow your coil to cool before you test again. Neat eh! Your plate should have shot up relatively quick!

ENHANCEMENTS

A fuse or circuit breaker will make things safer.

You could add a rheostat for fine levity control.

Technical Support Package

George C. Marshall Space Flight Center
Marshall Space Flight Center, Alabama 35812

NASA
National Aeronautics and
Space Administration

'Improved Acoustic Levitation Apparatus'

NASA Tech Briefs, Winter 1979
Vol. 4, No. 4, MFS-25050

METHOD AND APPARATUS FOR SHAPING AND ENHANCING ACOUSTICAL LEVITATION FORCES

ABSTRACT OF THE DISCLOSURE

A method and apparatus for enhancing and shaping acoustical levitation forces in a single-axis acoustic resonance system wherein specially shaped drivers and reflectors are utilized to enhance the levitation force and better contain fluid substances by means of field shaping.

Origin of the Invention

The invention described herein was made by employees of the United States Government and may be manufactured and used by or for the Government for governmental purposes without the payment of any royalties thereon or therefor.

Background of the Invention

This invention relates to an apparatus and method for suspending and positioning objects by the use of acoustical energy. In many instances, the study of materials and material processes and the determination of their properties can be more readily accomplished when the specimens under study are maintained out of contact with the walls of containers and solid supports. A levitated specimen of a material may be advantageously heated in a levitation furnace or by a spot heating device such as a laser.

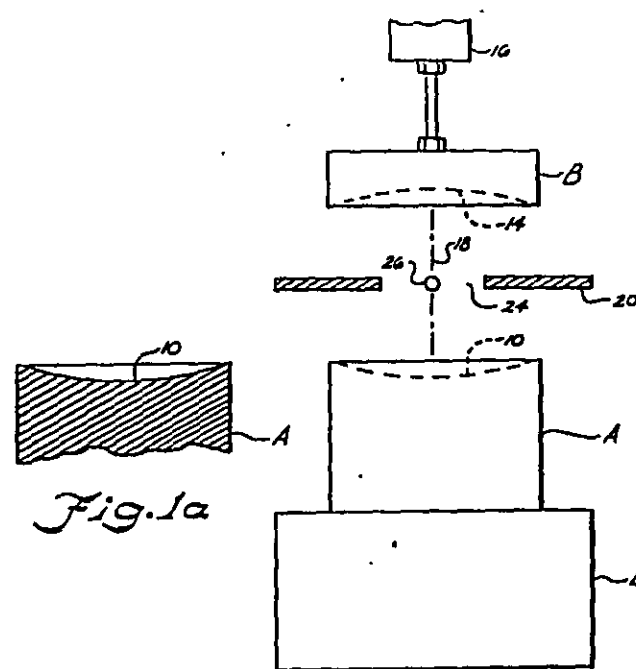


Fig. 1a

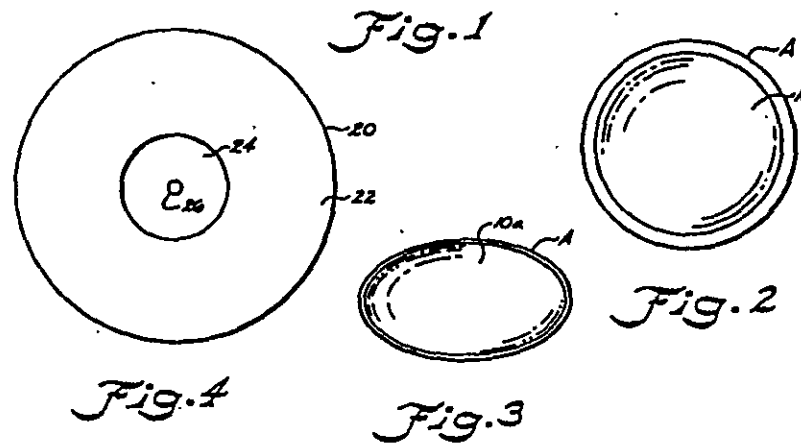


Fig. 4

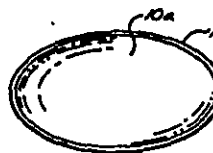


Fig. 3

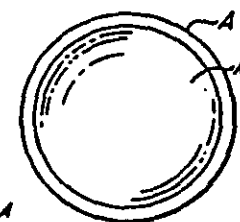


Fig. 2

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The fifth force: Pulling both ways

For 300 years physicists have believed that the force of gravity is the same for all substances. In Isaac Newton's words, it was the "quantity of matter," not the nature of the substance, that counted in determining the sizes of gravitational forces. Now some scientists have a highly controversial suspicion that the nature of the substance does matter (SN: 1/3/87, p.6).

The suspicion began with anomalies found in a series of measurements of gravity in mines and wells done in Australia several years ago. It is supported now by a new measurement done in a well in Michigan. Meanwhile, various laboratory experiments testing the earth's attraction for different chemical substances have come up positive and negative, the most recent being negative. As controversy continues, physicists are doing what they always do in such a situation — planning more experiments.

The Australians who did the original work, Frank Stacey of the University of Queensland in Brisbane and his collaborators, had found that in the mines, Newton's universal gravitational constant appeared slightly larger than the value determined by centuries of measurement in laboratories. To explain the discrepancy they suggested that a "fifth force" — or, as some have called it, a negative component of gravity — is at work. This would overthrow a lot of established physics, and the suggestion got a very negative reaction at first. It still does.

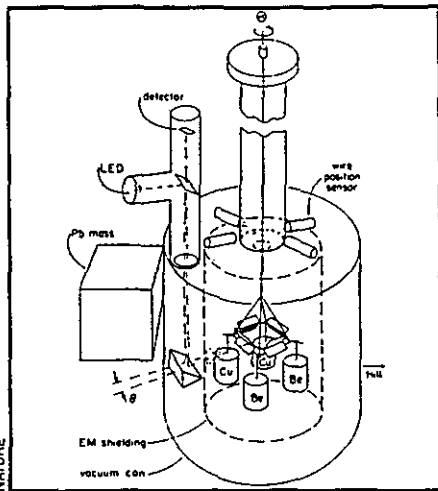
Now, however, a measurement done in a well in Manistee County, Michigan, in different geological and geographical surroundings, supports the Australian work. Albert T. Hsui of the U.S. Air Force Geophysics Laboratory in Bedford, Mass., and the Massachusetts Institute of Technology reports in the Aug. 21 *SCIENCE* that the Michigan work shows a similar apparent increase in the gravitational constant. As a possible explanation he quotes Stacey's suggestion of a fifth force and writes, "If the validity [of the work] can be confidently established, its implication for the laws of fundamental physics will be profound."

To be compatible both with these observations and with known physics, the fifth force would have to be medium-range, operating over distances between 10 and 1,000 meters or so. It would have to be 100 times weaker than ordinary gravity. It would be repulsive, and its strength would depend on the number of neutrons and protons in an object and so on the chemical elements present.

"Although a fifth force of this size certainly wouldn't change our everyday lives," says James E. Faller of the Joint Institute for Laboratory Astrophysics at the University of Colorado in Boulder, "it would have quite a profound effect on the

way we think about our world."

Interested physicists have been busy doing new experiments and checking to see whether the fifth force is compatible with the classic experiments of the past. Early in 1986, Ephraim Fischbach of Purdue University in West Lafayette, Ind., brought widespread attention to the fifth force by publishing a reanalysis of an experiment done 70 years ago by Roland von Eötvös (SN: 1/18/86, p.38). Until then physicists had regarded the Eötvös experiment as a demonstration that gravity takes no notice of the differences among substances. Fischbach reanalyzed Eötvös's data to show that they are compatible with the existence of the fifth force.



The Stubbs et al. experiment.

Galileo, back in the seventeenth century, also determined that gravity affects all substances in the same way. He is said to have done it by dropping balls off the leaning tower of Pisa. In the latest new experiment on this subject, Faller and graduate students Timothy M. Niebauer and Martin P. McHugh have redone the experiment in a modern guise. They dropped pieces of uranium and copper in evacuated towers — Faller calls them "20-centimeter leaning towers." (A vacuum is necessary to get rid of air resistance, which could affect different objects differently as they fall.) Mirrors on the weights allowed a laser to track their fall. The experimenters report in the Aug. 10 *PHYSICAL REVIEW LETTERS* that they found no difference in the earth's attraction for the two weights to an accuracy of 5 parts in 10^{10} .

Unlike Fischbach's result, theirs supports Galileo and traditional physics. With respect to the relative strength and range of a possible fifth force, they conclude, "This experiment extends the...region over which the usual gravitational laws are valid."

However, the record of the last year includes at least one positive result in an experiment of this sort. Peter Thieberger of Brookhaven National Laboratory in

Upton, N.Y., used a hollow copper sphere that he made neutrally buoyant so that it floated just submerged in water. He reasoned that, if the fifth force exists, the attraction of a large mass of earth for the copper will differ from the attraction of the same mass for the water, and the sphere should move in the water. Using a cliff of the New Jersey Palisades as an attractor, he found that the sphere did move. "This observation is consistent with a substance-dependent, medium-range force..." Thieberger concluded in the March 16 *PHYSICAL REVIEW LETTERS*.

The same journal also included a negative result by C. W. Stubbs and colleagues at the University of Washington in Seattle. In this case four samples, two of copper and two of beryllium, were arranged at the corners of a square and hung by a single wire in a torsion balance. If a large mass to the side of the apparatus (a hill on the campus) attracted copper and beryllium differently, the wire should have twisted. The negative results of the experiment, they conclude, "make it very unlikely that the geophysical and Eötvös anomalies have a common origin in a 'new force.'"

"It is always difficult to draw definite conclusions from such delicate experiments," comments E. Iacopini of the Scuola Normale Superiore in Pisa, Italy, reviewing the situation in the Aug. 13 *NATURE*. Iacopini describes two forthcoming experiments. In one of them, in which Iacopini will participate, a half-beryllium, half-copper sphere will fall in an evacuated chamber. If the earth's attraction for copper is different from its attraction for beryllium, the sphere will rotate somewhat about its axis as it falls. An arrangement of mirrors and lasers will measure that motion.

The second of these experiments, by Edoardo Amaldi of the University of Rome and his collaborators, involves the gravity-wave detector that belongs to the University of Rome and is set up at the CERN laboratory in Geneva, Switzerland. Gravity waves are cyclic disturbances of gravitational forces that propagate through space. They are gravity's analog to radio waves, and astrophysicists expect them to come from several kinds of astrophysical events. If they do, they should make the detector vibrate.

This experiment will attempt to excite the detector artificially. Near the detector, the experimenters will set up a "gravitational dipole," a hollow cylinder of which half is iron and half beryllium. This dipole will rotate at the detector's resonant frequency, 915 hertz. If the fifth force exists, the rotating dipole should excite the antenna, producing a signal that should be evident.

Summing up the situation, Iacopini remarks, "It is too early to draw any conclusions, but in a year or so there will be many experimental results."

— D. E. Thomsen

Force that counteracts gravity identified

NEW YORK (AP) — A new analysis of a classic 60-year-old physics experiment has produced surprising evidence of a previously unknown fundamental force that tends to weakly counteract gravity over short distances.

The peculiar force would allow a feather to fall to the ground faster than a lead brick, in the absence of wind resistance.

Four hundred years ago, Galileo disproved the common sense notion that heavier objects fall faster than lighter ones. Legend has it that he dropped two objects of different weights from the top of the leaning tower of Pisa. Both fell at the same rate.

The newly discovered force would slow the fall of the heavier object by a very small amount.

The existence of the new force, called the hypercharge force, was proposed in the Jan. 6 issue of Physical Review Letters, a prestigious physics journal, by Ephraim Fischbach of Purdue University in West Lafayette, Ind., and his colleagues — Daniel Sudarsky, Aaron Szafer and Carrick Talmadge from Purdue, and Sam Aronson from Brookhaven National Laboratory on Long Island, N.Y.

The researchers analyzed the re-

sults of the so-called Eotvos experiment.

In 1922, Roland von Eotvos and colleagues in Budapest, Hungary claimed to have confirmed Galileo's results in a classic experiment in which measurements of objects of different weights indicated that gravity acted equally on all of them.

Eotvos had blamed small errors in his measurements on the limitations of his equipment.

Fischbach and his group looked at those small errors and concluded that they were not errors, but were instead discrepancies that the scientists now attribute to the hypercharge force.

Four forces are now known in the universe: gravity, electromagnetism, and two sub-atomic forces called the strong force and the weak force. The hypercharge force would be the fifth.

Fischbach noted that he is not the first to find evidence of such a force. Frank Stacey of the University of Queensland in Brisbane, Australia, has during the last few years reported finding discrepancies in gravitational measurements. Fischbach now explains those discrepancies as evidence of hypercharge force.

"I myself think the evidence from all of the experiments is pretty convincing," Fischbach said Wednesday

from the University of Washington in Seattle, where he is on sabbatical leave from Purdue.

Some physicists who had not yet seen Fischbach's paper expressed skepticism over the result, but Robert Dicke of Princeton University, recognized as one of the world's leading authorities on gravitation, said he found the report "fairly convincing."

"I must say that with anything as revolutionary as this, you would like to see new and improved experiments," Dicke said. "I think it's a very important result if it's right. It tells us something about elementary particles, and it also requires a modification of our theories of gravitation."

Fischbach noted that the new force does not challenge Einstein's theory of gravitation. What Einstein said about gravity remains true, although Einstein said nothing about any other possible force like the hypercharge force.

Unlike gravity, which can act on objects hundreds of millions of miles apart, the hypercharge is believed to act only at distances of several thousand yards or less, Dicke said. Thus, it does not affect gravitational calculations of the movement of celestial bodies.

COMPRESSED AIR MAG.

creating the hypercharge analog of a magnetic field by spinning a heavy object in the vicinity of a test mass. What is exciting, of course, is that if such a force really exists, the technologies that could arise from this can hardly be anticipated at the present time." Unfortunately, editorial space restraints prevent us from expanding on Dr. Fischbach's comment.

The law of gravity, suggested by Galileo, formulated by Newton, and confirmed by Einstein, always did seem too straightforward. As every schoolboy suspected, the law is just too simple to last forever. Now we learn that a feather would fall faster than a coin if dropped from the same height in a vacuum. Scientists at Purdue University have confirmed a new force, called hypercharge, that acts upon falling objects of different composition in ways to cause them to accelerate at slightly different rates, not the same rate as previously accepted. Leading physicists have examined the preliminary findings of the researchers and, although not unanimous in their conclusions, tend to agree that the hypercharge factor could be potentially revolutionary and could necessitate the rethinking of many views of particle physics and cosmology. *Compressed Air* magazine contacted Dr. Ephraim Fischbach, who led the research group, to determine what immediate practical impact the findings might have in the everyday workings of our world. We learned from him that the hypercharge hypothesis impact lies mainly in the futuristic technologies it could spawn. Dr. Fischbach reports that "since this force can in fact be manipulated, we can anticipate the possibility of new technologies based on controlling this force in various ways. As an example, one can imagine

COMPRESSED AIR MAG. THE "LAW" OF GRAVITY TUMBLES AUG. 86

Scientists ponder existence of fifth fo

by JOHN NOBLE WILFORD
New York Times News Service

NEW YORK — A new analysis of early 20th-century experiments has produced results challenging both the findings of Galileo that all falling bodies accelerate at the same rate and a fundamental element of Einstein's general theory of relativity. That has led physicists to suspect there may be a fifth, heretofore unidentified force at work in the universe.

Scientists said the new study, published in the Jan. 6 issue of *Physical Review Letters*, could have a profound influence on thinking in physics and cosmology, if the results can be substantiated by further experiments. Those who had examined the report said it appeared to have been based on sound research.

Even though the new findings seemed to undermine a basic assumption made by Einstein and the principle of equivalence that stemmed from Galileo's work, scientists said the hypothesized new force, called the hypercharge, was so weak and local that, if it did exist, it should not fundamentally alter Einstein's principles as the basic tool of modern cosmology. The other known forces are electromagnetism, gravity and the strong and weak forces governing nuclear structure.

Feather falls faster

The new analysis suggests that, contrary to Galileo's assertion, a feather would fall faster than a coin if dropped from the same height in a vacuum. That is because, in the new thinking, gravity is not the only force at work; there is also presumably something called hypercharge, which acts on objects of different compositions in ways to cause them to accelerate at slightly different rates.

In a telephone interview, Dr. Ephraim Fischbach, the leader of the team of scientists who made the study, said: "When you see something as fundamental as a new force, it's likely to change many things. We will have to rethink many views of particle physics and cosmology."

Fischbach, a professor of physics at Purdue University in Indiana, is a visiting professor this year at the Institute of Nuclear Theory at the University of Washington in Seattle. The other authors of the report are Daniel Sudarsky, Aaron Szafer and Carrick Talmadge of Purdue and S.H. Aronson of the Brookhaven National Laboratory at Upton, N.Y.

Dr. Robert H. Dicke, a Princeton University physicist and authority on Einstein's theories, said: "One has to be somewhat careful when you're dealing with something that's potentially revolutionary. But if this is right, it's extremely important. I can't pick any

tain kinds of radioactivity.

The prospect of another fundamental force comes at a time when most theoretical physicists are laboring mightily to conceive a single mathematical framework that describes all the forces, excepting possibly gravity, as different manifestations of one general force. These concepts are called grand unified theories.

Dr. Heinz R. Pagels, president of the New York Academy of Sciences, who is the author of books on particle physics, noted that scientists had speculated about the possible existence of the hypercharge force for more than 20 years. "One remains skeptical until further experiments are done," he said.

What Fischbach and his team did was re-examine the data from experiments conducted by Roland von Eotvos, a Hungarian scientist, over a period of more than two decades and reported in 1922.

The experiments, involving the suspension of objects of different composition and mass from a torsion balance, appeared to confirm Galileo's observation in the early 17th century. In the familiar, though probably apocryphal, story, Galileo is supposed to have dropped cannon balls from the Leaning Tower of Pisa and thereby determined that all objects fall under gravity with the same acceleration, if differences attributed to wind resistance are ignored.

Newton relied on Galileo's work in formulating his theory of gravity in the late 17th century, and Einstein, reinforced by the Hungarian experiments, also made the assumption that all bodies fall at the same rate in a uniform gravitational field in enunciating his general theory of relativity in 1916.

However, according to Fischbach, even the Hungarian experimenters noted some discrepancies in their

results but chose to ignore being statistically insignificant. Upon detailed examination, Fischbach found the discrepancies in the group's "raw" experimental data to be large enough to suggest that some other force besides gravity might be acting on the bodies.

Fischbach was led to a review of the Eotvos experiment because of "funny things" — his term for the apparent effect of another force — that scientists had been seeing in recent years at the Fermi National Accelerator Laboratory in Illinois and in gravitational experiments by Frank Stacey deep in the mines.

In the report in *Physical Review Letters*, a journal of the American Physical Society, Fischbach concluded: "Although the experiment has been universally interpreted as having given null results, in fact that this is not the

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In the report in *Physical Review Letters*, a journal of the American Physical Society, Fischbach's group concluded: "Although the Eotvos experiment has been universally interpreted as having given null results, in fact that this is not the case." In other words, something seemed to be causing objects to fall at slightly different rates.

The group said, moreover, that the results supported the existence of an intermediate-range force, called hypercharge. Its strength may be as little as 1 percent of that of gravity, the weakest of the known forces, and its influence may extend little more than 600 feet. This is why, Fischbach said, the force is not likely to change calculations on planetary-system scales.

Hypercharge appears to be more like electromagnetism than gravity; in fact, the equations are similar. Both the Earth or any other large bodies and objects falling toward them are made up of matter in which the protons and neutrons have hypercharge in varying amounts depending on their composition. The hypercharge appears to work against gravity to affect the rate of acceleration.

Force of fifth force at work

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emission of light: the strong force binds the atomic nucleus together; and the weak force is responsible for cer-

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Geophysics on the fifth force's trail

Isaac Newton made one of the greatest discoveries in classical physics when he realized that the force that draws an apple toward the ground is the same one that keeps the moon in orbit around the earth. For hundreds of years, his formula for the gravitational attraction between two bodies has helped scientists and engineers calculate everything from the orbits of planets to the trajectories of rockets.

But gravity has given modern theorists a problem. As physicists have tried to combine all four known forces of nature — the gravitational, electromagnetic, weak and strong forces — into one unified field theory, they have been unable to incorporate gravity without postulating the existence of other, as-yet-undiscovered forces (SN: 7/26/86, p.55). In particular, the unified field theories say that the force of attraction between two bodies is given by Newton's formula plus a much smaller "fifth force" that seems to come into play at distances of about 100 to 1,000 meters.

Since this range is outside the scale of gravity measurements in the laboratory, theoretical physicists have turned to earth scientists to hunt for the fifth force. Indeed, a number of geophysical experiments have indicated such an effect, but because of experimental uncertainties they have yet to convince most scientists. A new generation of experiments is now in the works, and scientists reported on two of these in San Francisco at the recent meeting of the American Geophysical Union.

The basic purpose of the experiments is to measure the force of gravity at different elevations over several hundred meters and to compare these measurements with what Newton's formula predicts, based on the measured masses of nearby bodies. In practice, scientists measure an effective gravitational constant, which, in Newton's expression, relates the force of gravity to the masses and the distance between them.

One of the most comprehensive geophysical experiments so far has been conducted by Frank Stacey at University of Queensland in Brisbane, Australia, and his colleagues. Working in two metal mines, the researchers have measured a gravitational constant that is 0.7 percent greater than that measured in the laboratory — suggesting the presence of a fifth force. Other scientists, however, worry that Stacey's group has been unable to determine the density of surrounding rocks with sufficient precision. After making 14,000 boreholes into mine rocks, Stacey maintains that the mine density is well known, but he is worried that there may be some unknown, deeply buried bodies that are affecting his group's results.

So Stacey and others have been searching for geologic settings in which the densities and distribution of nearby masses are as uniform and well known as possible. Mark E. Ander at Los Alamos (N.M.) National Laboratory and Mark A. Zumberge, George E. Backus, Alan D. Chave, John Hildebrand and Fred N. Spiess of Scripps Institution of Oceanography in La Jolla, Calif., report that they are planning to do an experiment next summer in a 2,000-meter-deep borehole in the Greenland ice sheet. The researchers believe they will be able to measure the gravitational constant to better than 1 part in 1,000 over depths of about 100 to 1,500 meters down the borehole. To achieve this accuracy, they are taking painstaking care in their measurements. For example, they will "season" or stretch the wire line, which will hold the gravity meter, many times before the experiment and will calibrate its absolute length in an Idaho silver mine both before and after their experiment.

Ander's group would like to repeat the ice experiment in a Soviet borehole in Antarctica. This group and, independently, Stacey's group are also planning experiments in the ocean from about 100 to 3,000 meters' depth. Measuring depths is easier in the ocean than in a borehole, and the seafloor topography is more even than the rocky terrain underlying ice. However, no one has measured gravity through the water column to such depths before, so there are technical problems still to be worked out.

Another experiment was proposed at the meeting by Christopher Jekeli and Andrew R. Lazarewicz at the Air Force Geophysical Laboratory at Hanscom Air Force Base, Mass. They want to measure gravity up along one of two several-hundred-meter-tall television towers in Houston or in Raleigh, N.C. The advantage of this approach, says Jekeli, is that the effect of the air density is far less important than that of water, ice or rock. Stacey's group has tried a similar experiment, but it did not succeed because the tower was shaking too much. Jekeli and Lazarewicz, however, think their tower will be stable enough.

Jekeli and Ander say their experiments are not likely to be definitive: If they don't find evidence for a fifth force, the theorists can respond that the force must act over greater distances than the experimental ranges, and if they do measure a gravitational constant different from the Newtonian value, skeptics can always say this result was due to experimental errors.

In this respect, these are high-risk experiments, says Ander. But over time, such experiments may yield a big payoff. What's most exciting about doing this work, he says, is that "there are very few opportunities for geophysicists to make an impact on fundamental physics, and this is clearly one of them." — S. Weisburd

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NEWS

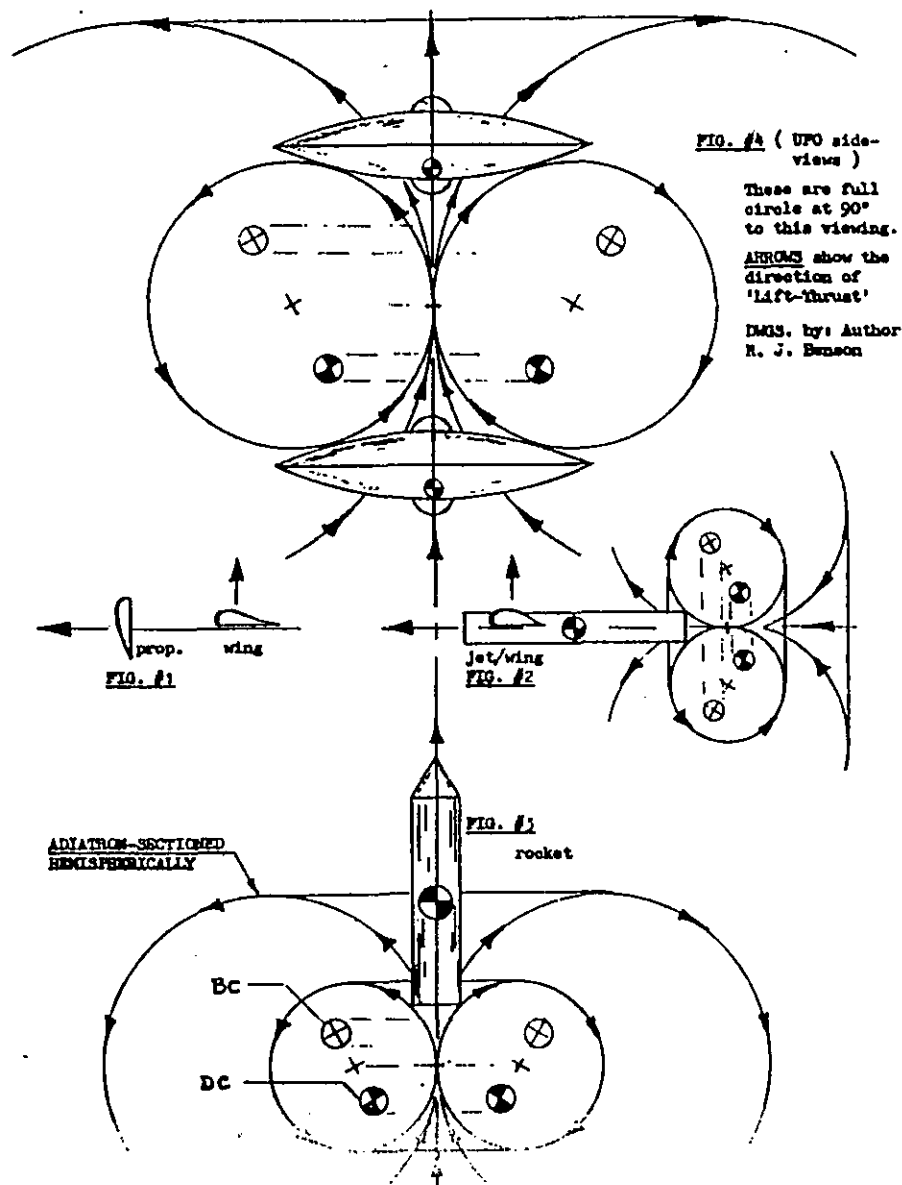
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by: Richard J. (Dick) Benson 1977

UFO's & OTHER CRAFT's ADIATRNIC-FIELD of PROPULSION



Introduction:

THRUST & LIFT as currently used aerodynamically, aerostatically, and even hydrodynamically are the same phenomenon termed herein as the ADIATRNIC-FIELD effect as currently used by Air, Sea and Space craft though each employs that single phenomenon to differing-degrees of efficiency as thus requires a proportional amount of fuel consumption for mobility.

In nature, the ADIATRNIC-FIELD is evidenced clearly as the mechanics of motion (vertical) within all adiabatic conditions such as that of the dust-devil, the cyclone, hurricane, tornado, explosion, etc., except for the flight-mode we do not have the cyclonic-horizontal-rotation-effect.

Adiatronic-Fields all teach identical relative form as may be considered generally spherical and outfolding omni-radially at the head-zone of it's axis of motion and infolding omni-radially at the rear-zone of that axis. These fields of force (atmospheric, electro-magnetic, etc.) all expand ahead and away from the leading aspect of it's axis and thus relatively contract inward towards the trailing aspect of it's axis of mobility. The form is somewhat likened to a spherical doughnut wherein the axis of the doughnut hole is it's axis of mobility. Because of Adiatron is generally displacing more mass of the medium it passes through it exhibits a relative buoyancy (LIFT) force-expression....this is the force as may be used to express direction when issuing from a craft since the craft will be pushed and pulled by that field-effect as natural equilibration of force balancing as the craft literally 'sails' into the low-pressure zone in reacting to it's relative 'suction' and simultaneously is 'blown' rearward as rushes in to fill that relative void of the low-pressure zone.

The 'ARROWS' shown in all FIGURES of the drawings teach the direction of LIFT-THRUST motion of the Adiatronic-field as well as of the rotary-wing (propeller) and fixed-wing craft, and of the rocket-type and UFO-type craft.

FIG. #1 Our 'least-efficient' aircraft use the rotary-propeller-wing and fixed-wing arrangements. The flatter-surface (rear of prop/bottom of wing) pushes against the air to 'plane' the craft in the opposite direction in assisting it to move into the low-pressure-zone created adjacent it's curved-surface (ahead of prop/over wing) which pulls that member proportionally, and the industry evaluations show that two-thirds the LIFT-THRUST of flight is due to the low-pressure side of the reaction of equilibration of the forces. The propeller-velocity of spin dictates the forward-motion of the craft as well as it's altitude-ceiling acquisition capability in cancelling one against the other which established the 'Service-Ceiling' for optimal-efficiency. Size of the airfoils (prop & fixed wings) are limited by inherent inefficiencies, hence the least LIFT-THRUST is the rule for these type craft.

FIG. #2 Our Jet-Engine aircraft though "super-sonic" operate at about 2.25x the speed-of-sound via issuing rearward a vast ADIATRNIC-SYSTEM of expanding gas-exhaust which through it's omni-radiating expanding-head zone pushes the craft ahead and thus forward in horizontal flight inline with it's axis of motion. This we allude to the Newtonian 'law of motion' as relatively 'equal-and-opposite' reaction. Simultaneously this push of the craft activates it's fixed-wing to assist stabilization of the body of the craft along the axis of the Adiatron. Obviously the velocial gain of the craft over the propeller-type is due to the greater LIFT-THRUST capacity of the considerably-sized adiatronic-field rearward.

FIG. #3 Our Rocket-Propulsion Space-Craft as employ 25,000-mph 'escape-velocity' for interplanetary flight via ballistics-trajectory through space, use the greatest amount of fuel for 'lift-off' and to reach 'escape-velocity' and during this series of 'burns' it's issue rearward develops our largest ADIATRNIC-FIELD utilization for flight-craft and as may be seen by the nuclear-explosion when

to merely 'hover' in any position and with application of 'trim' forces the craft could hold-against wind-effects over any given location of a surface.

'All Flight' of the UFO-Type craft is the product of natural-vectoring-effect in the equilibration of all forces attendant the function as that portion of the maneuver which without evident surface attachments finds the craft capable of efficient motion in any-direction it's pilot choses.

The drawings herewith show a bi-convex lens-shaped UFO-Type craft
since this shape appears optimally suited for all conditions of
power-on and power-off maneuverability with reasonable safety. The
Adiatronic-Field coupled with the relative 'neutral-specific-density'
manipulation with respect to the craft's mass-weight finds then that
the craft could be made to efficiently fly in any shape. ie: A cube,
sphere, ellipse, rectangle, cone, pyramid, etc., though these may not
accord them the 'planning-efficiency' in any-direction as does the
bi-convex lens shape of this drawing.

Mass-weight-limitation of the UFO-Type craft is virtually limitless, though the larger displacement craft would probably not enter our lower atmospheric levels due to pressure-differentials as may induce unwanted stress for that size craft. We are thus probably visited from time to time by the smaller sized craft which can operate independently of, or be from a larger unit in-space.

NOTES: The Adiatronic-Field proportions of these sketches are disproportionate in showing them considerably smaller than reality to fit them onto the page. The field envelopes the craft totally during most modes of flight. Even the jet-aircraft and the rocket are within their adiatronic-fields along the axis of it's mobility at velocital acquisition. The field in reaching us at ground-level from the jet and rocket cause the sound-barrier shock-wave, and the UFO's efficiency finds it's field near ground-level as being barely perceptible to our hearing senses.

While some UFO-types may use exotic-fuel, it is a fair assumption that they all predominantly use 'solar-energy' as they can re-direct it during flight while relatively not holding onto much, if any, of it for any extended period. Within Earth-Planet celestial envelope the average velocity of 'light' is 675-Million-mph wherein "LV" = $21.384/1 + \pm 21.384$ as = .675-Billion-mph when:

4	= 31.68	& •	31.56	& 675-Million mph = x1000-
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31.68	= 21.384/.675	&	31.68-Mega-Seconds = Solar-Year (366-Days)	
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4.01 = Diametric & Hemispheric-Plane-Area proportion of Sphere,
      'REED-CUBIT'
      (1-Surface-Area / 4.1 = .3168-Hemispheric-Plane-Area)

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      (ie: 3.156-GIRT x .3168-REED = 1 as 1/.3168 = 3.156 &
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'In-Flight-Powering-Down' by the pilot of the UFO-type or allows the craft to "cut itself" 'plane' into up or down position or

3168 ROOT = E as = 2.01 of the drawing

31.56 = TIME-ROOT in mega-seconds-CALEND-Year. (86,400-sec-day)

31.68 = TIME-ROOT in mega-seconds-UNIVER-Year. (86,400-sec-day)

.005 290 56 = 1-RA Corner (90°) Face of E²-BOR-E (E²G.25/4 = 1-RA Corn)

.016 727 032 = BALL volume as E³/E6 = B & as BxExE = E³ (1/x funct.)

.031 794 75 = CUB-E as = E³ volume

.044 600 988 = HYP-L as = hypotenuse- ANGLE-FACES (both as shown on dwg.)

.125 = .5ARE-B at 'Sea-Level' of water-surface equalized orbitally.

.1584 = BOR-E ARE.5 as 3.156 x .3168 x.5 (GxEx.5)

.25 = ARE-B as total 'Sea-Level' BALL AREA = 100-Billion-ACRES-#

.3168 = BOR-E ARE as total I.D. Face of bore of dwg.

.416^k = ARC-KNOP (BULL) off Ecliptic-Plane for "SUN" measuring.

.78914 = BOR-E AREA-end-portion of E²G.25 & = .5B-Plane-AREA

.81^k = ARC-KNOP (BOSS) as .0264 SUN-DISK-TRIANGULATION UNIT-ARC

1. 'Integer' = UN-I for all study-portion-unit-whole: Ball, Arc, Arc, Cub, Bor, etc. (is: .3168E/3168E = 1G circle-area of BALL as UNI-G, 1-CUB = E 'square', and CUB/G.25 = 1-BOR-E and CUBG.25=CILI)

1.4444 = HIP-L (Hypotenuse-line) .45758595 = Ex 1.4444 of RA-54^k/90°

3.156 = UNI-G root for GIRIN as 3.156 x E = 1 DIAMETER-INTEGER

31.68 = .5BALL-PLANE = 31.68-BILLION-ACRE-SURFACE @ 43,560-Sq-Ft-Each

63.36 = STABLE (Statute-ACRE-BALL-Levels .5E 'Back-To-Back') CUBIT.

.1-STABLE-MIL (Statute-Mile = 63,360-I-per-SMILE) = 1-STADIA

72. = NAUTE-CUBIT via .864-factor(mega-sec-day x 10) 63.36 / .864 = 72. Naute-cubit-Integer. 25,000-SMILE x .864 = 21,600-NMILE & 60-NMILE or 69.4-SMILE = 1° Arc-Compass or 1.6° Arc, and 36-NMILE or 41.6-SMILE = 1^k Arc-Compass or .6° Arc equally.

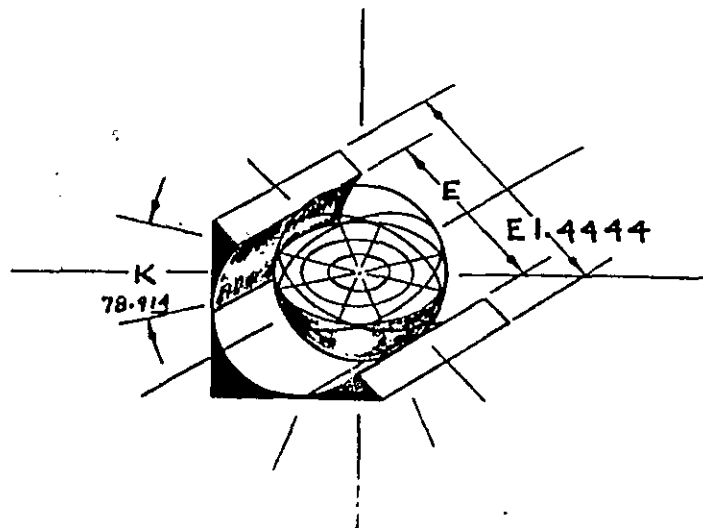
78.914 = TROPIC-ZONAL-ARC per .5AN(CALEND-YEAR of 31.56-Mega-Seconds) = .5-TIME-ROOT as 7.6032 xG.25 = 600-Knop-Compass(360°)

100. = BULL-ACRE-BALL as = ARE-B "UNITY" SURFACE-AREA of EARTH-BALL

3168. = CHORD as 'LINE-MEASURE' of TROPIC-ZONAL-ARC perpendicular to Ecliptic-Plane-Vertical and intersecting the Surface-Arc in Statute-Miles (SMILES) wherein the surface Arc = 3288.098 and the vertical-Secant-axis = 501.8112-SMILE-measure-Integer.

7920 = SMILE (Stable-MIL-E diametric-value) for EARTH-BALL-UNITY at this factor-magnitude of dimensioning-measure-proportion integer use & Chord-Secant's 501.8112-SMILE x 15.782 = 7920-SMILE-DIAMETER.

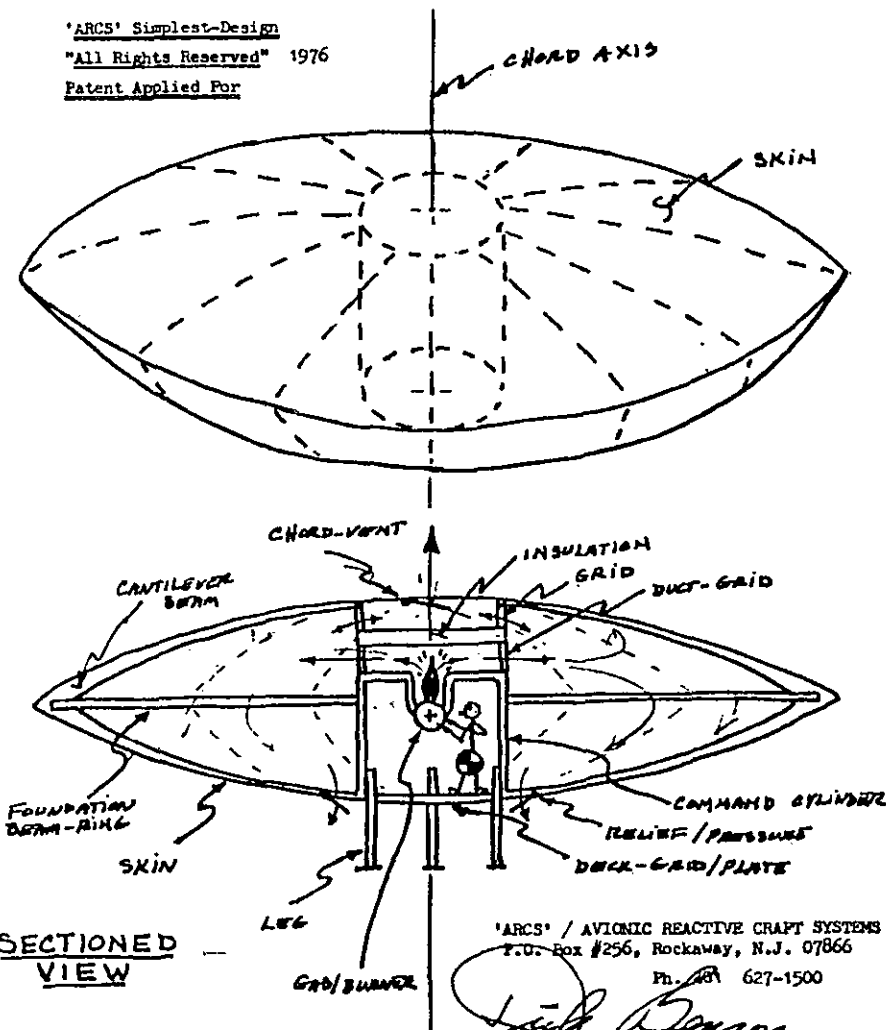
25000 = UNI-G-SMILE value for EARTH-BALL & x E (.3168) = 7920-Diameter.



'ARCS' Simplest-Design

"All Rights Reserved" 1976

Patent Applied For



SECTIONED
VIEW

'ARCS' / AVIONIC REACTIVE CRAFT SYSTEMS
P.O. Box #256, Rockaway, N.J. 07866
Ph. 609 627-1500

Richard J. (Dick) Benson

DIAMAGNETIC FIELD GENERATION AND UTILITY--CLARK COILS

The attached design for an electric coil-levitator shows one way to generate a diamagnetic strong field levitator. If four or six of these coils were mounted on the board and the unit turned upside down, the coil-board platform would levitate from the earth as a floating platform because the earth is basically diamagnetic. Of course the levitator unit will levitate wood, fish, and stones, all of which are diamagnetic materials. If these sound familiar, they should, for they are Schauburger's logging flumes, trout up the waterfalls and floating/impacting "Rhinegold" stones. The multiple coil strong levitating platform is the V-7 Project Fighter Aircraft (flying saucer).

Let us wind an electrical D.C. type coil levitator out of hollow copper tubing. If we pass a large flow of electric current through the copper tube we will generate a strong magnetic field and a fairly strong diamagnetic levitator field. Now we pump only water at maximum flow rate through this coil and generate levitation effects (diamagnetism without a measurable magnetic field). Thus, a helical water flow coil (Clark Coil) produces diamagnetic effects directly without the production of a magnetic field. Generally any helical (coil) flow produces some diamagnetism and in electric current flows it also produces a strong magnetic field. This comingling of diamagnetism with electrical effects has hidden the true nature and utility of diamagnetism from us.

Now, with the coil flow as water only, repeat Oersted's generator experiment. Loop some wire as a coil in a complete circuit through an ammeter and move this coil back and forth through the strong purely diamagnetic field at close range. No magnetism, but electric current flows in the moving coil circuit. What produces current flow in any electric generator? Magnetism or so-called "diamagnetism". An electric or permanent magnet is a crude and inefficient method of producing "diamagnetic" fields to generate electrical flow (generators). Also, electric/permanent magnets are crude and inefficient methods of producing repulsive levitational effects for lifting platforms (vehicles) or missile launching weapons (firearms). We need to design high flow water pumps and directional flow water coils (Clark Coils) to build massive Over-Unity Electric Generators cheaply, levitating vehicles, and powerful firearms.

In the Over-Unity Generators designed on water flow Clark Coil generated dismagnetism, the radiating diamagnetic central water coil and the wire field coils are stationary--only a light conical field shield rotates between them for electrical generation. Of course, just electronically completing the field coil circuits would make a motionless generator with apparent motion output far over unity. Power output to power input (pump/rotary drive) is in the thousands of percent because we are now doing it the right way.

With massively powerful "diamagnetic" fields from the coils, the lifting or levitating against the earth as vehicles (platforms) is easy. Who needs highways or airports for a transport system? And by vectoring some of the lift forces horizontal movement is achieved at rapid rates.

The diamagnetic firearms follow naturally from these powerful repulsive fields--far superior to any electrically coil produced devices. These weapons have no recoil or noise. Only air resistance and friction limit the velocity that can be achieved. Electrostatically charged rounds may greatly increase the terminal velocity of these weapons. An energy or mechanical shutter could be used for firing.

Remember in building your water coils (Clark Coils) to wind them as D.C. electrical coils are wound. Start at one end and coil tightly downward; at the bottom return gently to the top in less than one rotation around the coil and wind another complete layer downward. This makes the field directional and additive--maximum strength. Far better Clark Coil designs are possible to increase power per unit flow.

A few minutes thought well show how Searl and Brown made crude diamagnetic generators. The Faraday Unipolar (N Machine) generators are crude devices for the same reason. Also we are at the heart of the Kervran/Kushi element transmutations and radioactive halflife control.

BE CAREFUL--All of these devices levitate.

Technical Notes: The Clark Coil that was built in the above data is a fairly powerful 10^{12} Hz (Spark Discharge or Zero Point) radiation generator unit. This frequency (10^{12} Hz) is nature's force field for various atomic changes, bindings, and field power transfers. As an example of atomic level control, current flow in a conductor (any electrical generator) can be viewed as a weak, partial Kervran/Kushi atomic element transmutation. Every current conduction electric generator, conventional or Over-Unity, functions on this principle. This is because we have started to strip the orbital electrons, to bare the nuclei for fusion or fission by the application of a weak source of 10^{12} Hz power to a held diamagnetic material. As an example of levitators/propellers, the so-called diamagnetic atomic element samples react by moving directly away from a strong local source of this 10^{12} Hz radiation if free to do so. This movement is because these element samples are structurally conservative, balanced and stable systems within the total local energy environment. The Solar System is a big model of this levitator system. Incidentally, the ferrous metals react to 10^{12} Hz as any other element when no magnetic field is present to interfere.

The unfortunate misnaming and misunderstanding of this one energy field system under various aspects as diamagnetic, nuclear or gravity fields has hidden its true nature and utility from us until now. The frequency 10^{12} Hz is generated by spark discharge, vacuum (zero point), certain types of water flows, the sun, etc., and only to a much lesser degree by permanent or electromagnets and capacitors. The true nature, functioning, and utility of this energy field is largely undeveloped to date due to hastily made early assumptions which concreted into dogma.

Academic science to date has created at least four names for this same force field and its various effects. The names created to date are gravity, strong nuclear, weak nuclear, and diamagnetic fields. Nature only has the three force fields and they are the electrical, magnetic, and gravity (10^{12} Hz). I have chosen the oldest historical name, gravity, for this field energy (10^{12} Hz) and it thus includes all of the other names by which this field has been called over the years.

Nature functions in elegant triads--two powerful antithetical elements meet at and generate a "neutral center" element. Each of these three primary elements has two antithetical extremes and a null or "neutral center" structure forming triads also. There are only three primary force fields, three whole primary particles, and three dimensions in nature's triadically designed universe.

In order to make a critical judgment of competing theories one must compare what current academic establishment physics predicts and constructs with what my theory predicts and constructs. What this means is do you want the currently offered crude atomic control, under-unity generators, fuel gulping limited vehicles and noisy recoiling weapons of establishment science or do you want vastly over-unity generators and levitation vehicles and weapons?

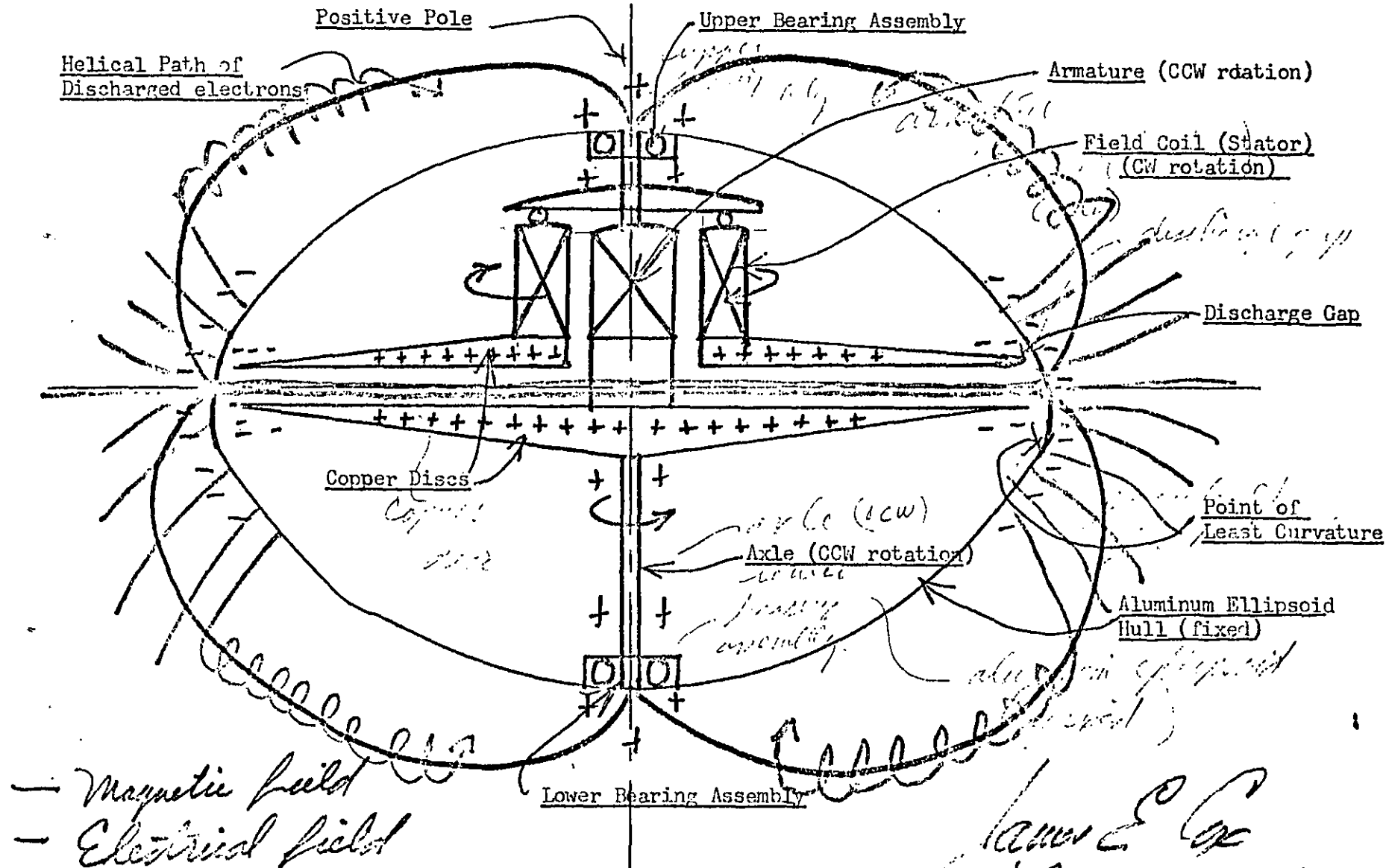
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Richard LeFors Clark, Ph.D.
March 18, 1986

SPECULATIVE DESIGN OF GYROMAGNETIC
SYSTEM EMPLOYING THE SEARL EFFECT



SPECULATIVE DESIGN OF GYROMAGNETIC
SYSTEM EMPLOYING THE SEARL EFFECT

EXPLANATION OF DRAWING:

This particular embodiment of the GYROMAGNETIC PROPULSION SYSTEM consists of a hollow one-piece aluminum ellipsoid which contains a pair of copper discs, each fixed to the armature and field coil of an electrical motor such that the discs counter-rotate about each other in opposite directions. The only external energy input is the current supplied to the motor, which is attached conductively to the hull and receives its current via slip rings and brushes not shown in the attached drawing.

OPERATION OF DEVICE:

Copper is characterized by the fact that it has a relatively large number of free or mobile valence electrons, which makes it an excellent conductor. Because of this mobility, forces may act on these electrons to cause motion and thereby change the electron charge distribution. Thus, when the motor is actuated in the device depicted in the attached drawing, centrifugal forces act upon the electrons and force them to move outwardly towards the periphery of the spinning copper discs. The interior or central portion of the disc, and its attached axle thereby becomes positively charged; while the rim becomes negatively charged. Experiments of Michael Faraday showed that the charge on a closed metal surface resides on the outside of the surface, there being no net potential difference within the surface or ellipsoid in this case. He further showed that electrons tend to discharge at points of least curvature where the electrical field intensity tends to be strongest.

Aided by the centrifugal forces of the spinning copper discs, the electrons discharge from the rim of the discs and jump over to the outer portion of the aluminum hull — where they set up an electrostatic field. The moving, now positively charged copper discs each set up a magnetic field around themselves that oppose each other, causing the discs to tend to repel away from each other magnetically, as well as statically due to their like charges. If electrons start to discharge from the rim of the aluminum hull, they will move in a helical path in the magnetic field generated within the hull by the rotating copper discs and these electrons will be "captured" by the positively charged pole of the entire system. A current flow is thus set up — moving outwards from the center of the discs to the rim, across the discharge gap onto the aluminum hull where they collect and again discharge into the magnetic field which directs them to the pole of the system where they are "taken in" to complete the circuit.

The voltage potential is obviously a function of the rotational speed of the discs as well as the nature of the material used in the discs. When a certain threshold speed is reached, or voltage potential, the system becomes weightless and thrust is generated (SEARL EFFECT). Can this be explained by orthodox physical principles?

ENCLOSURE #2

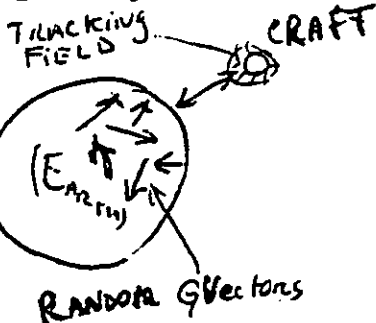
SIGNED
DATED

James E. G.
23 November 1969

TRIDEL RESEARCH

ORIGINAL REPORT DATE: 01-22-80

The correlational tie ~~between~~ between alignment and randomness verses gravity:



Gravity is mathematically represented most easily as a "Pink Noise" frequency(ies) of a given power output directly traceable to the majority elemental masses present in the planet's (or any structure) construction. For a given system this can be generally shown as a vectoral randomness using a Bessel curve but the limits of the equation are set to the atomic density of the element in predominance and the quantity present per unit volume.

An Anti-G vector-noise medium must have elastic non-cohesion to itself but be able to be held within confines such as the walls of a balloon. Hot air achieves relativistic (used loosely) alignment due to high thermal agitation allowing a greater incidence of non-planetary bucking alignment with the individual masses on the planet.

A better medium than hot air would be helium because of its ability to reorient itself atomic spin-wise with the gravity vectors coming off the individual and as a whole atoms of the planet.

Optimumly sufficient non-bucking alignment must occur in the vectoral directions of the earth or G-mass to balance the structure into an anti-grav situation. Mass only becomes evident when one bucks the flow.

This phenomena is directly applicable to all forms of study.

All the craft's G-vectors are thereby placed into a constantly tracking aligned state and cause no resistance to the vectors randomly (loosely) being generated by the larger mass (the planet.) The ratio in the craft structure is called the "Packing Factor".

Mass defined: In a given reference system there are interreacting vectors having rays which are generated by electron spin. The quantity of these vectors and the density of the randomness of emanation /incidence direction and the vectoral direction of a disturbing ray(rays) determines the quantity of resistance experienced by the disturbance. This is called MASS. The greatest mass occurs when, in a simple example for explanation, a singularity is aligned in a clockwise spin and another singularity of equal charge (E,H,G,) rotating counterclockwise tries to overtake. The resultant effect is infinite mass or the imovable object. Zero mass for the same conditions would occur through simultaneous rotation in the same alignment plane.

As velocities increase if nothing is done about randomness, relativistic mass increases. Align moving reference frame with vectoral direction will allow no apparent increase in mass seemingly violating Einstein's laws on relativity.

Simply, faster than light travel can be achieved by treating the universe as a random "Pink Noise" frequency and using the above mechanics. R.Dratch

TRIDEL RESEARCH

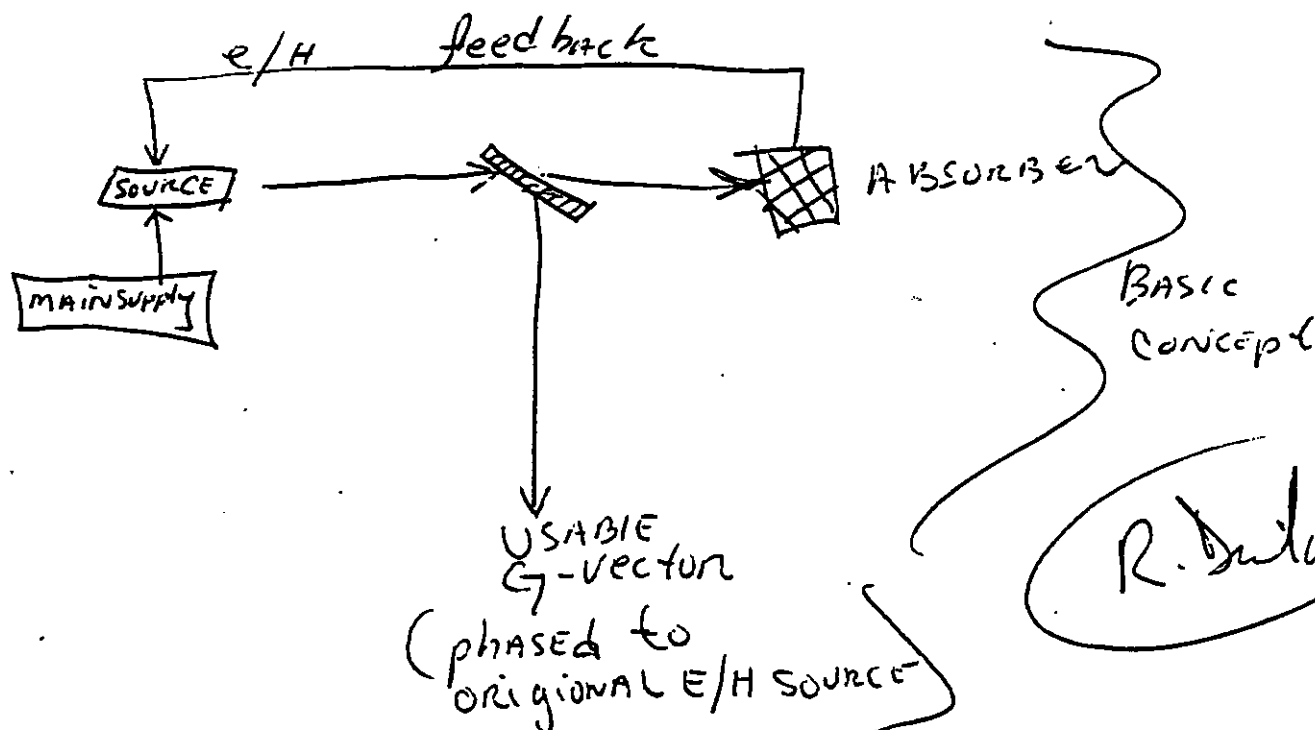
ORIGINAL REPORT DATE: 11-06-79

Notes on the electromagnetic-gravitational phenomenon-R. Dratch w/h. Blair

1. The E&H vectors balanced out will cause G to return to point of origin or T-initial. There will be no apparent time lag in doing so.
 2. Changing one vector in energy amplitude will cause a corresponding change in either of the other two. The system will try to achieve balance in the most easy manner.
 3. If a half silvered reflector is placed incident to an E,H,G, wavefront generally the electromagnetic wave constituents will pass through while the G or Time Related transport component will be distorted via the E,H, structures of the reflector so as to be somewhat reflected.
- Note that there is a necessity that the reflective-transmissive characteristics (due to the energy density, and randomness coefficient) of the half-silvered reflector match the energy/frequency and alignment consideration of the incident E-H component so that transmission of E-H and not blockage occurs.
4. In referring to three above, G then can be used as long as the source remains constant and E-H are directed away from the G-vector area.
 5. G then can be isolated, combined with new systems and thereby allow new systems to come into existence.

6. G isolated then can be manipulated so that space-time-gravity warps in continuum can be achieved.

7. Mode Hops are more easily achieved.



TRIDEL RESEARCH

ORIGINAL REPORT DATE: 25 August 1979 R.Dratch
GENERAL LAW OF G "FIELDS":

It is the dissipation of energy of the G vector via the intermediates of the E and H creation which generally tie together nearspace phenomena. G or time/existence is a product of the quantity of X-Space, or Zero crossings of an electromagnetical alternating field usually consisting as a multiple spectral waveform, or signature.

In balancing thermodynamic laws, a balanced or steady state is achieved. This is most easily done through system alignment to an excitation field close to the natural resonant frequencies of the matter.

If balance is altered the system "RINGS" through overshooting the steady state. This shows the elasticity or the ability of a medium to partially achieve alignment before interatomic cohesion takes over. *With elasticity added to the mass.*

This elasticity factor is directly related to the cross-vectors of the intra-atomic makeup per unit area of given substance. A condition called resonance occurs when the impinging vectoral wave is in sympathy with the elasticity/cross-vector frequency of the medium.

Note that heat changes resonant frequencies due to elasticity and cross-vectoral density or alignment changes.

Fractional studies of matter indicate cross-vectoral densities of varying intensities. These generally track on electron shell filling and quantity. Correlation can be had by impinging a ray of force to the sample and noting the deformations and relaxation/elasticity times. Fields used in this manner will create closed loop crystalline domains within the substance.

Harmonically related alignment conditions can occur which can lead to similar characteristics in atomic elemental constituents. Extrapolating, universal parallels can exist if alignment on all 3 vectors can be had, even if only harmonically phase related.

If E&H have a ringing effect than why not G? (law of balances)
In general, overlapping fields of reference systems create distortions which thereby become perceivable to an incident wavefront measuring device. Reality manifests through offering "Resistance" to perception.

A spark provides the initial disturbance of equilibrium through shear space E,H,G overlapping and overload. It is through resistance to alignment in a vector situation that solidification occurs. It is through closing the vector ray into a loop that perpetuation occurs.

Standing waves and node shifting through an impulse seems as a proper energy system to close loops.